ANTI-CANONICAL MOVING SESHADRI CONSTANTS AND VARIETIES OF FANO TYPE

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Abstract. We prove that a normal projective variety is of Fano type if the moving Seshadri constant of the anti-canonical divisor is at least the dimension of the variety. We thus answer Question 8.2 of Zhuang's paper on Fano varieties with large Seshadri constants.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Seshadri constants, introduced by Demailly [Dem92b], measure the local positivity of a nef divisor at a point. Bauer and Szemberg [BSz09, Theorem 1.7] showed that if X is a smooth Fano variety of dimension d and $\epsilon(-K_X, x) > d$ for some point $x \in X$, then $X \cong \mathbb{P}^d$. Murayama [Mur18, Theorem B] showed that $X \cong \mathbb{P}^d$ for a smooth Fano variety X of dimension d over an algebraically closed field of any characteristic with $\epsilon(-K_X, x) \geq d+1$ for some point $x \in X$. For a $\mathbb{Q}$-Fano variety Z of dimension d (Definition 2.2) with a smooth point z such that $\epsilon(-K_Z, z) > d$, we have $Z \cong \mathbb{P}^d$ [LZ18, Theorem 2]. The $\mathbb{Q}$-Fano varieties Z of dimension d with a smooth point z such that $\epsilon(-K_Z, z) = d$ are classified in [LZ18, Theorem 3], and we recall the list in Theorem 7.2.

Zhuang [Zhu18] removed the assumptions on the singularities of X by replacing the Seshadri constant with the moving Seshadri constant, introduced by Nakamaye [Nak03]. For a normal projective variety X and a smooth point $x \in X$, the moving Seshadri constant $\epsilon_{\mathrm{mov}}(-K_X, x)$ is the asymptotic growth rate in m of the largest order of jets at x separated by $H^0(X, \mathcal{O}_X(-mK_X))$, that is, the limit superior as $m \to +\infty$ of s/m, where s is the largest integer such that $H^0(X, \mathcal{O}_X(-mK_X))$ separates s-jets at x, and $\epsilon_{\mathrm{mov}}(-K_X)$ is the supremum of $\epsilon_{\mathrm{mov}}(-K_X, x)$ over all smooth points $x \in X$ (Definition 2.4). When X is smooth and $-K_X$ is big, for any point $x \in X$, $\epsilon_{\mathrm{mov}}(-K_X, x)$ coincides with the moving Seshadri constant of $-K_X$ at x in the sense of Nakamaye [ELM+09, Proposition 6.6]. In [Zhu18] the moving Seshadri constant is denoted by ϵ_m. Zhuang asked whether the bound $\epsilon_{\mathrm{mov}}(-K_X) \geq d$ alone forces X to be of Fano type (Definition 2.2).

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Question 1.1 ([Zhu18, Question 8.2])**.** Let X be a normal projective variety of dimension d such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d$. Is X of Fano type?

A positive answer to Question 1.1, together with [LZ18, Theorem 3], lists all normal projective varieties X of dimension d with $\epsilon_{\mathrm{mov}}(-K_X) = d$ up to isomorphism in codimension one [Zhu18, Introduction, the second paragraph after Theorem 1.8]. The threshold d cannot be lowered: for any integer $d \geq 2$ and any real number $\eta > 0$ there exists a Fano variety X of dimension d with worse than log canonical singularities, hence not of Fano type (since K_X is $\mathbb{Q}$-Cartier, a klt pair (X, Δ) with $\Delta \geq 0$ would make $(X, 0)$ klt by [KM98, Corollary 2.35(1)]), such that $\epsilon_{\mathrm{mov}}(-K_X) > d - \eta$ [Zhu18, Theorem 1.8]. Here $-K_X$ is ample, and the moving Seshadri constant of an ample divisor is the usual Seshadri constant [Zhu18, Section 1, equation (1) and the paragraph after it] (equation (1) there concerns line bundles at a general point, and for the $\mathbb{Q}$-Cartier divisor $-K_X$ one uses it for a Cartier multiple of $-K_X$ together with Lemmas 2.8 and 2.9 and the fact that $\epsilon(-K_X, x)$ attains its maximum over the smooth points x at a very general point [Zhu18, Section 1, the paragraph after Definition 1.1]).

Question 1.1 has a positive answer for X in the following cases.

(1) $\epsilon_{\mathrm{mov}}(-K_X) > d$, in which case $X \cong \mathbb{P}^d$ [Zhu18, Theorem 1.7].
(2) $\epsilon_{\mathrm{mov}}(-K_X) = d$ and $\mathrm{vol}(-K_X) > d^d$ [Zhu18, Theorem 1.8(1) and Corollary 4.1].
(3) $d = 2$ and $\epsilon_{\mathrm{mov}}(-K_X) = d$ [Zhu18, Theorem 1.8(2)].
(4) $\epsilon_{\mathrm{mov}}(-K_X) = d$, and there exists a $\mathbb{Q}$-divisor $D \geq 0$ such that $-(K_X + D)$ is $\mathbb{Q}$-Cartier and nef and the Seshadri constant of $-(K_X + D)$ at some smooth point of X is at least d [Zhu18, Theorem 1.8(3)].
(5) $d \geq 3$, $\epsilon_{\mathrm{mov}}(-K_X) = d$, and there exists a birational map $g : X \dashrightarrow \mathbb{P}^d$ such that g^{-1} does not contract any divisor [Zhu18, Theorem 1.10].

To obtain [Zhu18, Corollary 4.1], Zhuang constructs an isolated non-klt center at a very general point $x \in X$ using a $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -K_X$ and $\mathrm{mult}_x D = d(1 + t)$ for some small rational number $t > 0$, and the existence of D is where the inequality $\mathrm{vol}(-K_X) > d^d$ is used. When $\epsilon_{\mathrm{mov}}(-K_X) = d$, counting jets only shows $\mathrm{vol}(-K_X) \geq d^d$. Indeed, for any real number $0 < \eta < d$, $H^0(X, \mathcal{O}_X(-mK_X))$ separates jets of order at least $(d - \eta)m$ at some smooth point for any m in an infinite set M of positive integers, so $h^0(X, \mathcal{O}_X(-mK_X)) \geq \binom{\lceil (d-\eta)m \rceil + d}{d}$ for any $m \in M$ and $\mathrm{vol}(-K_X) \geq (d - \eta)^d$. If $d = 1$, then X is a smooth projective curve, and $\epsilon_{\mathrm{mov}}(-K_X) \geq 1$ forces $\deg(-K_X) > 0$, since otherwise no non-zero section of $\mathcal{O}_X(-mK_X)$ vanishes at a point. Thus $X \cong \mathbb{P}^1$ and $\epsilon_{\mathrm{mov}}(-K_X) = 2 > d$, since $H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(2m))$ separates $2m$-jets and no $(2m + 1)$-jets at any point, so X is covered by [Zhu18, Theorem 1.7]. Therefore the cases of Question 1.1 not covered by [Zhu18, Theorem 1.7 and Theorem 1.8(1)(2)] are exactly those with

$$d \geq 3, \quad \epsilon_{\mathrm{mov}}(-K_X) = d, \quad \text{and} \quad \mathrm{vol}(-K_X) = d^d.$$

We now explain how far the methods of [Zhu18] go when $d \geq 3$, $\epsilon_{\mathrm{mov}}(-K_X) = d$, and $\mathrm{vol}(-K_X) = d^d$. For a klt variety X of dimension d with $\epsilon_{\mathrm{mov}}(-K_X) = d$, Zhuang constructed a $\mathbb{Q}$-Fano variety Z, a smooth point $z \in Z$ with $\epsilon(-K_Z, z) \geq d$, and a birational map $g : X \dashrightarrow Z$ such that g^{-1} does not contract any divisor [Zhu18, Proposition 1.9]. If $\epsilon(-K_Z, z) > d$, then $Z \cong \mathbb{P}^d$ [LZ18, Theorem 2]. If $\epsilon(-K_Z, z) = d$, then Z is one of the varieties in Theorem 7.2 [LZ18, Theorem 3]. The existence of g alone does not imply that X is of Fano type, and Zhuang has two ways to conclude from the variety Z.

(1) If $\mathrm{vol}(-K_Z) = d^d$, then comparing volumes by [Zhu18, Lemma 2.15] shows that $a(E, Z) \leq 1$ for any prime divisor E on X contracted by g [Zhu18, Remark 5.5], and X is of Fano type by [Zhu18, Lemma 2.9]. Zhuang regards this case as the direct one [Zhu18, Introduction, the paragraph after Proposition 1.9].
(2) If there exists a birational map $X \dashrightarrow \mathbb{P}^d$ whose inverse does not contract any divisor, then X is of Fano type by [Zhu18, Theorem 1.10]. Such a map $X \dashrightarrow \mathbb{P}^d$ exists when $Z \cong \mathbb{P}^d$ and when Z is a blow-up of $\mathbb{P}^d$.

In the list of Theorem 7.2, the quartic weighted hypersurfaces in $\mathbb{P}(1^d, 2, 2)$ and the quotients of quadrics satisfy $\mathrm{vol}(-K_Z) = d^d$ (Lemma 7.3) and are treated by the first way, the blow-ups of $\mathbb{P}^d$ along the complete intersection of a hyperplane and a hypersurface are treated by the second way, and the log del Pezzo surfaces occur only when $d = 2$. Neither way works directly for the weighted hypersurfaces

$$Z = (x_0 x_{d+1} = f) \subset \mathbb{P}(1^{d+1}, r),$$

where $x_0, \dots, x_d$ have weight 1, x_{d+1} has weight r for an integer $r \geq 1$, and $f \in \mathbb{C}[x_1, \dots, x_d]$ is homogeneous of degree $r + 1$. On one hand, by Lemma 4.2(4),

$$\mathrm{vol}(-K_Z) = \frac{d^d(r+1)}{r} > d^d = \mathrm{vol}(-K_X),$$

and comparing volumes leads to no contradiction. On the other hand, let $o = [0 : \cdots : 0 : 1] \in Z$. The blow-up of Z at o is the blow-up of $\mathbb{P}^d$ along $(x_0 = f = 0)$, and the exceptional divisor Γ of the blow-up of Z at o is the strict transform of the hyperplane $(x_0 = 0)$ [Zhu18, Section 5, item (1)]. Thus the inverse of the projection $Z \dashrightarrow \mathbb{P}^d$ from o contracts the hyperplane $(x_0 = 0)$ to o, and the inverse of the composite $X \dashrightarrow Z \dashrightarrow \mathbb{P}^d$ does not contract any divisor if and only if the valuation of Γ is the valuation of a prime divisor on X. For example, if $d = 3$, $r = 1$, and $f = x_1^2 + x_2^2 + x_3^2$, then $Z = (x_0 x_4 = x_1^2 + x_2^2 + x_3^2) \subset \mathbb{P}^4$ is a smooth quadric threefold with $(-K_Z)^3 = 54$, whereas $\mathrm{vol}(-K_X) = 27$, and the inverse of the projection $Z \dashrightarrow \mathbb{P}^3$ from o contracts the plane $(x_0 = 0)$ to o. After [Zhu18, Theorem 1.10], Zhuang remarks that the case when Z is such a weighted hypersurface seems more involved and that he does not see how to proceed.

For the usual Seshadri constant, Zhuang's positive-characteristic sequel proves the analogous statement by reduction modulo p: if $\Delta \geq 0$ is a $\mathbb{Q}$-divisor on X such that $-(K_X + \Delta)$ is $\mathbb{Q}$-Cartier and ample and $\epsilon(-(K_X + \Delta), x) \geq d$ for some smooth point $x \in X$, then X is of Fano type [Zhu22, Corollary 1.4]. Moving Seshadri constants do not occur in [Zhu22, Corollary 1.4]. The hypothesis of Question 1.1 provides no such Δ, since it only controls the asymptotic order of the jets separated by $H^0(X, \mathcal{O}_X(-mK_X))$, and K_X need not be $\mathbb{Q}$-Cartier. For the $\mathbb{Q}$-Fano variety Z given by [Zhu18, Proposition 1.9] and $\Delta = 0$, [Zhu22, Corollary 1.4] shows only that Z is of Fano type. However, [Zhu18, Lemma 2.8] carries the Fano type property from X to Z, and not from Z to X.

In this paper we answer Question 1.1 completely: the answer is positive in any dimension.

Theorem 1.2. *Let X be a normal projective variety of dimension $d \geq 1$. Assume that $\epsilon_{\mathrm{mov}}(-K_X) \geq d$. Then X is of Fano type.*

Sketch of the proof. By Proposition 3.2 and Lemma 3.1, Theorem 1.2 reduces to Theorem 7.1 on $\mathbb{Q}$-factorial terminal varieties Y with $d \geq 3$, $\epsilon_{\mathrm{mov}}(-K_Y) = d$, and $\mathrm{vol}(-K_Y) = d^d$. By [Zhu18, Proposition 1.9], there exists a birational map $g : Y \dashrightarrow Z$ such that g^{-1} does not contract any divisor, and by [LZ18, Theorem 2], Theorem 7.2, Theorem 2.11, Lemma 7.3, and Proposition 7.4, Y is of Fano type unless Z is a weighted hypersurface $(x_0 x_{d+1} = f) \subset \mathbb{P}(1^{d+1}, r)$. Theorem 6.1 treats the weighted hypersurfaces using a log canonical threshold τ on Z of the anti-canonical sections of Y (Construction 5.2). If $\tau > 1$, then a general anti-canonical member of Y yields a klt boundary (Lemma 6.2). If $\tau \leq 1$, then $\tau = 1$ and, after a change of coordinates by Lemma 4.5 if $r = 1$, ord_Γ computes τ (Proposition 6.3 and Lemma 4.3), and extracting Γ over Y reduces Theorem 6.1 to [Zhu18, Theorem 1.10] (Lemma 6.6). The key step is Proposition 5.3: if the anti-canonical sections of Y vanish along a valuation v with $a(v, Z) < +\infty$ to asymptotic order $\sigma(v) > 0$ (Construction 5.2), then Nadel vanishing implies

$$\sigma(v)^d \,\mathrm{vol}(v) \leq \mathrm{vol}(-K_Z) - \mathrm{vol}(-K_Y) = \frac{d^d}{r},$$

and d^d/r is the minimum of normalized volumes (Definition 2.13) at o, attained only by multiples of ord_Γ, while the minimum at smooth points is d^d (Lemma 4.3).

Only the weighted hypersurfaces need new inputs: Proposition 4.8, Propositions 5.3 and 6.3, and Lemma 6.6. The proof shows that a klt boundary exists, but does not decide whether $\tau > 1$ or $\tau = 1$.

The proof of Theorem 6.1 uses four new ideas.

(a) Boundaries from the anti-canonical sections of Y. Applied to g and the log Fano pair $(Z, 0)$, [Zhu18, Lemma 2.9] needs $a(F, Z) \leq 1$ for any g-exceptional prime divisor F on Y. For the weighted hypersurfaces, some F satisfies $a(F, Z) > 1$, since otherwise any section of $-kK_Z$ with k divisible enough would be a section of $-kK_Y$, and $\mathrm{vol}(-K_Y) \geq \mathrm{vol}(-K_Z) > d^d$, contradicting $\mathrm{vol}(-K_Y) = d^d$. For a general $u \in H^0(Z, \mathcal{O}_Z(-kK_Z))$, the pair $(Z, (\mathrm{div}_Z(u) - kK_Z)/k)$ is klt, but $\mathrm{div}_Y(u) - kK_Y$ has coefficient $k(1 - a(F, Z)) < 0$ along F. We therefore only use the sections $u \in H^0(Y, \mathcal{O}_Y(-kK_Y))$, viewed on Z (Construction 5.2). For such u, the pairs $(Y, (\mathrm{div}_Y(u) - kK_Y)/k)$ and $(Z, (\mathrm{div}_Z(u) - kK_Z)/k)$ have equal log discrepancies, since both log canonical divisors are $1/k$ times the divisor of the rational function u.

(b) Centers of valuations. In [Zhu18, proof of Theorem 1.10, the paragraph after display (2)], if the anti-canonical sections of degree k, viewed on $\mathbb{P}^d$, vanish to order λk at a point of the center of a valuation, then the moving Seshadri constant of the anti-canonical sections is at most $d + 1 - \lambda$, which is $(-K_{\mathbb{P}^d} \cdot \ell) - \lambda$ for a line $\ell \subset \mathbb{P}^d$ through that point, and $d + 1 - \lambda < d$ only if $\lambda > 1$.

On Z, let p_Z be a very general point. On the chart $(x_0 \neq 0) \cong \mathbb{A}^d$, the closure D_p of the cone $(f = 0)$ translated to have vertex p_Z is an ample Cartier divisor (Construction 4.6 and Lemma 4.7(1)), and the curves C of Lemma 4.7(2) through p_Z, which satisfy $(-K_Z \cdot C) = d$ and so work for any $\lambda > 0$, are contained in D_p. We use D_p itself: it intersects any positive-dimensional closed subvariety, avoids o, and contains the singular points of Z other than o (Lemma 4.7). By Izumi's theorem [Izu85, Ree89] and since $\epsilon_{\mathrm{mov}}(-K_Y) = d$ (Lemma 5.4), the center of a prime divisor E over Z along which the anti-canonical sections of Y vanish to order linear in the degree is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$ (Proposition 4.8).

(c) A local bound by Nadel vanishing. The comparison of volumes [Zhu18, Lemma 2.15] only yields $\mathrm{vol}(-K_Y) < \mathrm{vol}(-K_Z)$ and does not say at which point of Z the anti-canonical sections of Y lose the difference d^d/r. Since no vanishing theorem is available for their base ideals on Z, we use the asymptotic multiplier ideals $\mathcal{J}_k$ of Construction 5.2. By (b), the center q of a valuation v with $\sigma(v) > 0$ is an isolated point of the support of $\mathcal{O}_Z/\mathcal{J}_k$ for $k \gg 0$ (Lemma 5.6). By Nadel vanishing (Lemma 2.17(1)(2)), the length of $\mathcal{O}_Z/\mathcal{J}_k$ at q is at most the codimension of $H^0(Y, \mathcal{O}_Y(-kK_Y))$ in $H^0(Z, \mathcal{O}_Z(-kK_Z))$, and Lemma 2.17(4) bounds the length from below by the colength of a valuation ideal of v. Comparing leading coefficients in k yields Proposition 5.3. Since [Blu18, Theorem B.1] only provides a real valuation (Definition 2.12) computing τ, (b) and Lemma 2.17(4) are proved for real valuations (Lemma 5.5).

For $Z \cong \mathbb{P}^d$, Nadel vanishing would only bound $\sigma(v)^d \,\mathrm{vol}(v)$ by

$$\mathrm{vol}(-K_{\mathbb{P}^d}) - \mathrm{vol}(-K_Y) = (d+1)^d - d^d,$$

which exceeds the minimum d^d of normalized volumes at any point of $\mathbb{P}^d$, so it would not rule out $\tau < 1$.

(d) Extracting Γ without losing sections. If Γ is not a divisor on Y, then we extract Γ over Y in order to use [Zhu18, Theorem 1.10] (Lemma 4.4). A projective birational morphism $Y' \to Y$ whose only exceptional divisor is Γ keeps a section u of $-kK_Y$ if and only if $\mathrm{mult}_\Gamma(\mathrm{div}_Y(u) - kK_Y) \geq k(a(\Gamma, Y) - 1)$, so it may lose sections and lower the moving Seshadri constant. Moreover, [BCHM10, Corollary 1.4.3] extracts Γ from a klt pair (Y, Δ) with $a(\Gamma, Y, \Delta) \leq 1$, whereas $a(\Gamma, Y) > 1$ as Y is terminal. By Proposition 6.3 and Lemma 6.4, the asymptotic order $\sigma_Y(\mathrm{ord}_\Gamma)$ of the anti-canonical sections of Y along Γ (Construction 5.2) equals $a(\Gamma, Y)$. Thus any section of $-kK_Y$ vanishes along Γ to order at least $ka(\Gamma, Y)$, hence survives on Y'. Since $\tau = 1$, for a rational number $c < 1$ close to 1, some k, and a general member G of $|-kK_Y|$, the pair $(Y, cG/k)$ is klt, and $a(\Gamma, Y, cG/k) \leq (1 - c)a(\Gamma, Y) < 1$ (Lemma 6.6).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce moving Seshadri constants of Weil divisors, normalized volumes of real valuations, and asymptotic multiplier ideals of graded sequences of ideals. We also recall results of Zhuang [Zhu18, Lemma 2.8 and Theorem 1.10] and results on log canonical thresholds of graded sequences of ideals [Blu18, Section 2.9, Theorem B.1, and Proposition B.2].

In Section 3 we settle the cases $\epsilon_{\mathrm{mov}}(-K_X) > d$ and $\mathrm{vol}(-K_X) > d^d$ using [Zhu18, Theorem 1.7 and Corollary 4.1], the case $d = 2$ using [Zhu18, Theorem 1.8(2)], and the case $d = 1$ directly. In the remaining case we pass to a $\mathbb{Q}$-factorial terminal modification $Y \to X$, whose push-forward map $H^0(Y, \mathcal{O}_Y(-kK_Y)) \to H^0(X, \mathcal{O}_X(-kK_X))$ is an isomorphism for any positive integer k, and reduce Theorem 1.2 to showing that a $\mathbb{Q}$-factorial terminal projective variety Y of dimension $d \geq 3$ with

$$\epsilon_{\mathrm{mov}}(-K_Y) = d \quad \text{and} \quad \mathrm{vol}(-K_Y) = d^d$$

is of Fano type.

In Section 4 we study the weighted hypersurfaces $Z = (x_0 x_{d+1} = f) \subset \mathbb{P}(1^{d+1}, r)$. We show that if linear subspaces $V_k \subset H^0(Z, \mathcal{O}_Z(dk))$ vanish to order at least kt along a prime divisor E over Z for some $t > 0$ and separate jets of order asymptotically at least dk at a very general point, then the center of E on Z is a smooth closed point or the point $[0 : \cdots : 0 : 1]$.

In Section 5 we consider a birational map $g : Y \dashrightarrow Z$ to such a weighted hypersurface Z with klt singularities such that g^{-1} does not contract any divisor. For any positive integer m such that $-mK_Z$ is Cartier, we regard $H^0(Y, \mathcal{O}_Y(-mK_Y))$ as a linear subspace of $H^0(Z, \mathcal{O}_Z(-mK_Z))$, and for any valuation v of $\mathbb{C}(Z)$ with finite log discrepancy along which the graded linear series $\big(H^0(Y, \mathcal{O}_Y(-mK_Y))\big)_m$ has positive asymptotic order of vanishing, we show that the center of v on Z is a closed point and bound the asymptotic order from above in terms of the volume of v and the difference $\mathrm{vol}(-K_Z) - \mathrm{vol}(-K_Y)$.

In Section 6 we show that Y is of Fano type if there exists a birational map $g : Y \dashrightarrow Z$ to such a weighted hypersurface Z with klt singularities such that g^{-1} does not contract any divisor.

In Section 7 we treat the other varieties in the classification of [LZ18, Theorem 3], using [Zhu18, Proposition 1.9], [LZ18, Theorem 2], [Zhu18, Theorem 1.10], and a comparison of volumes as in [Zhu18, Remark 5.5], and prove Theorem 1.2.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs and varieties of Fano type.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\,\mathrm{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a normal projective variety. We say that X is *of Fano type* if there exists a $\mathbb{Q}$-divisor $\Delta \geq 0$ on X such that (X, Δ) is klt and $-(K_X + \Delta)$ is ample. A *$\mathbb{Q}$-Fano variety* is a normal projective klt variety X such that $-K_X$ is ample.

Varieties of Fano type are defined in [PS09, Lemma-Definition 2.6].

For a birational map $g : X \dashrightarrow X'$ between normal projective varieties, the condition that g^{-1} does not contract any divisor means that any prime divisor on X' is the birational transform of a prime divisor on X. The condition that g^{-1} does not contract any divisor is equivalent to the condition in [Zhu18, Definition 2.7], and any projective birational morphism $X \to X'$ satisfies it.

Theorem 2.3 ([Zhu18, Lemma 2.8])**.** *Let $g : X \dashrightarrow X'$ be a birational map between normal projective varieties such that g^{-1} does not contract any divisor. Assume that X is of Fano type. Then X' is of Fano type.*

2.2. Moving Seshadri constants.

Definition 2.4. Let X be a normal projective variety and D a Weil divisor on X.

(1) For a smooth point $x \in X$ and a linear subspace $V \subset H^0(X, \mathcal{O}_X(D))$, we let $s(V, x)$ be the largest integer $s \geq 0$ such that the evaluation map $V \to \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1}$ is surjective, and we let $s(V, x) := -1$ if the evaluation map $V \to \mathcal{O}_X(D) \otimes \mathcal{O}_X/\mathfrak{m}_x$ is not surjective.

(2) For a sequence $V_\bullet = (V_m)_{m\geq 1}$ of linear subspaces $V_m \subset H^0(X, \mathcal{O}_X(mD))$ and a smooth point $x \in X$, we let
$$\epsilon_{\mathrm{mov}}(V_\bullet, x) := \limsup_{m\to+\infty} \frac{s(V_m, x)}{m}.$$
We say that $V_\bullet$ is a *graded linear series* of D if $V_m \cdot V_l \subset V_{m+l}$ for any $m, l \geq 1$, where $H^0(X, \mathcal{O}_X(mD))$ is identified with $\{h \in \mathbb{C}(X)^* \mid \mathrm{div}(h) + mD \geq 0\} \cup \{0\}$.

(3) The *moving Seshadri constant* of D at a smooth point $x \in X$ is
$$\epsilon_{\mathrm{mov}}(D, x) := \epsilon_{\mathrm{mov}}(V_\bullet, x),$$
where $V_m := H^0(X, \mathcal{O}_X(mD))$ for any $m \geq 1$. We let
$$\epsilon_{\mathrm{mov}}(D) := \sup_{x\in X_{\mathrm{reg}}} \epsilon_{\mathrm{mov}}(D, x).$$

In Definition 2.4, $\epsilon_{\mathrm{mov}}(V_\bullet, x)$ is defined for any sequence $V_\bullet$ of linear subspaces: $V_\bullet$ need not be a graded linear series, and some V_m may be 0. We use such sequences for subspaces cut out by vanishing conditions along valuations (Proposition 4.8) and for the images of anti-canonical sections under a birational map (Construction 5.2). We use the reflexive sheaves $\mathcal{O}_X(mD)$ for any $m \geq 1$, not only for m divisible by a Cartier index, because K_X need not be $\mathbb{Q}$-Cartier.

By Lemma 2.9, when $V_\bullet$ is a graded linear series, the limsup in Definition 2.4 does not change if m runs only over the multiples of a fixed positive integer I, and hence, when K_X is $\mathbb{Q}$-Cartier, $\epsilon_{\mathrm{mov}}(-K_X, x)$ is also the limsup over the multiples of the Cartier index of K_X only.

Definition 2.5. Let X be a normal projective variety of dimension d and D a Weil divisor on X. The *volume* of D is
$$\mathrm{vol}(D) := \limsup_{m\to+\infty} \frac{d!h^0(X, \mathcal{O}_X(mD))}{m^d}.$$

Theorem 2.6 ([FKL16, Theorem 3.5(i)])**.** *Let X be a normal projective variety of dimension d and D a Weil divisor on X. Then*
$$\mathrm{vol}(D) = \lim_{m\to+\infty} \frac{d!h^0(X, \mathcal{O}_X(mD))}{m^d}.$$
In particular, $\mathrm{vol}(ND) = N^d \mathrm{vol}(D)$ *for any positive integer N.*

We recall the Seshadri constant of a nef divisor (cf. [Zhu18, Definition 1.1]).

Definition 2.7. Let X be a normal projective variety, L a nef $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and $x \in X$ a smooth point. Let $h : X' \to X$ be the blow-up of X at x and E the exceptional divisor of h. The *Seshadri constant* of L at x is

$$\epsilon(L, x) := \sup\{t \in \mathbb{R}_{\geq 0} \mid h^*L - tE \text{ is nef}\}.$$

We say that a property holds at a *very general point* of X if it holds at any point of the intersection of countably many dense open subsets of X. We include a proof of the following lemma (cf. [Zhu18, Section 3], where the conclusion appears without proof in the paragraph before Lemma 3.1).

Lemma 2.8. *Let X be a normal projective variety, D a Weil divisor on X, and $V_\bullet = (V_m)_{m\geq 1}$ a sequence of linear subspaces $V_m \subset H^0(X, \mathcal{O}_X(mD))$. Then there exists a subset $U \subset X_{\mathrm{reg}}$ that is the intersection of countably many dense open subsets of X such that $s(V_m, x) \geq s(V_m, y)$ for any $x \in U$, any $y \in X_{\mathrm{reg}}$, and any $m \geq 1$. In particular,*

$$\epsilon_{\mathrm{mov}}(V_\bullet, x) = \sup_{y \in X_{\mathrm{reg}}} \epsilon_{\mathrm{mov}}(V_\bullet, y)$$

for any $x \in U$, and $\epsilon_{\mathrm{mov}}(D) = \epsilon_{\mathrm{mov}}(D, x)$ for a very general point $x \in X$.

Proof. Let $\dim X = d$. Fix $m \geq 1$ and an integer $s \geq 0$. Let $W \subset X_{\mathrm{reg}}$ be a non-empty open subset such that $\mathcal{O}_X(mD)|_W \cong \mathcal{O}_W$ and there exist regular functions $z_1, \ldots, z_d$ on W such that $dz_1, \ldots, dz_d$ is a basis of Ω_W. Such open subsets W cover X_{reg}. Let $\partial_1, \ldots, \partial_d$ be the derivations of $\mathcal{O}_W$ dual to $dz_1, \ldots, dz_d$, and write $\partial^\alpha := \partial_1^{\alpha_1} \cdots \partial_d^{\alpha_d}$ for $\alpha \in \mathbb{Z}^d_{\geq 0}$. Let $\sigma_1, \ldots, \sigma_N$ be a basis of V_m, regarded as regular functions on W via the trivialization of $\mathcal{O}_X(mD)|_W$. For $y \in W$, Taylor expansion in $z_1 - z_1(y), \ldots, z_d - z_d(y)$ shows that the evaluation map $V_m \to \mathcal{O}_X(mD) \otimes \mathcal{O}_X/\mathfrak{m}_y^{s+1}$ is given by the matrix

$$M_s(y) := \left(\partial^\alpha \sigma_i(y)\right)_{1 \leq i \leq N,\ |\alpha| \leq s}.$$

Thus the evaluation map is surjective if and only if $M_s(y)$ has rank $\binom{d+s}{d}$, that is, some minor of $M_s(y)$ of order $\binom{d+s}{d}$ is non-zero (if $N < \binom{d+s}{d}$, the map is never surjective and the set below is empty). Since surjectivity at order s implies surjectivity at any order $s' \leq s$, the set $\{y \in W \mid s(V_m, y) \geq s\}$ is open in W. Hence $\{y \in X_{\mathrm{reg}} \mid s(V_m, y) \geq s\}$ is open in X.

If $s(V_m, y) = s \geq 0$, then V_m surjects onto $\mathcal{O}_{X,y}/\mathfrak{m}_y^{s+1}$, whose dimension is $\binom{d+s}{d} \geq s+1$. Thus $s(V_m, y) \leq \dim V_m - 1$ for any $y \in X_{\mathrm{reg}}$, and

$$s_m := \max_{y \in X_{\mathrm{reg}}} s(V_m, y)$$

exists. The set

$$U_m := \{y \in X_{\mathrm{reg}} \mid s(V_m, y) \geq s_m\} = \{y \in X_{\mathrm{reg}} \mid s(V_m, y) = s_m\}$$

is a non-empty open subset of X, and it is dense since X is irreducible. Let $U := \bigcap_{m \geq 1} U_m$. Then $s(V_m, x) = s_m \geq s(V_m, y)$ for any $x \in U$, any $y \in X_{\mathrm{reg}}$, and any $m \geq 1$. Since $\mathbb{C}$ is uncountable, X is not a countable union of proper closed subsets, so $U \neq \emptyset$.

Let $x \in U$ and $y \in X_{\mathrm{reg}}$. Since $s(V_m, x)/m \geq s(V_m, y)/m$ for any $m \geq 1$, we have $\epsilon_{\mathrm{mov}}(V_\bullet, x) \geq \epsilon_{\mathrm{mov}}(V_\bullet, y)$. Since $x \in X_{\mathrm{reg}}$, the equality

$$\epsilon_{\mathrm{mov}}(V_\bullet, x) = \sup_{y \in X_{\mathrm{reg}}} \epsilon_{\mathrm{mov}}(V_\bullet, y)$$

follows. Taking $V_m = H^0(X, \mathcal{O}_X(mD))$ for any $m \geq 1$, we obtain $\epsilon_{\mathrm{mov}}(D) = \epsilon_{\mathrm{mov}}(D, x)$ for any $x \in U$. □

Lemma 2.9. *Let X be a normal projective variety, D a Weil divisor on X, $V_\bullet$ a graded linear series of D, and $x \in X$ a smooth point. Then $s(V_{qm}, x) \geq q s(V_m, x)$ for any positive integers q and m. In particular,*

$$\epsilon_{\mathrm{mov}}((V_{Im})_{m\geq 1}, x) = I \epsilon_{\mathrm{mov}}(V_\bullet, x)$$

for any positive integer I.

Proof. Let $\dim X = d$, let q, m be positive integers, and let $s := s(V_m, x)$. If $s = -1$, then $qs \leq -1 \leq s(V_{qm}, x)$. Thus we may assume that $s \geq 0$.

Since x is a smooth point, D is Cartier near x. Let $u \in \mathbb{C}(X)^*$ be such that $D = \operatorname{div}(u)$ near x. Then the stalk of $\mathcal{O}_X(lD)$ at x is $u^{-l}\mathcal{O}_{X,x}$ for any $l \geq 1$, and for any integer $j \geq 0$ the evaluation map $V_l \to \mathcal{O}_X(lD) \otimes \mathcal{O}_X/\mathfrak{m}_x^{j+1}$ is identified with the map $V_l \to \mathcal{O}_{X,x}/\mathfrak{m}_x^{j+1}$ sending σ to the class of σu^l. Let $t_1, \ldots, t_d$ be a regular system of parameters of $\mathcal{O}_{X,x}$, and write $t^\gamma := t_1^{\gamma_1} \cdots t_d^{\gamma_d}$ and $|\gamma| := \gamma_1 + \cdots + \gamma_d$ for $\gamma \in \mathbb{Z}^d_{\geq 0}$.

Since $V_m \to \mathcal{O}_{X,x}/\mathfrak{m}_x^{s+1}$ is surjective, for any $\beta \in \mathbb{Z}^d_{\geq 0}$ with $|\beta| \leq s$ there exists $\sigma_\beta \in V_m$ such that

$$\sigma_\beta u^m = t^\beta + r_\beta, \quad r_\beta \in \mathfrak{m}_x^{s+1}.$$

Let $\gamma \in \mathbb{Z}^d_{\geq 0}$ with $|\gamma| \leq qs$. Since $|\gamma| \leq qs$, we may write $\gamma = \beta_1 + \cdots + \beta_q$ with $\beta_j \in \mathbb{Z}^d_{\geq 0}$ and $|\beta_j| \leq s$ for any j. Let $\tau_\gamma := \sigma_{\beta_1} \cdots \sigma_{\beta_q}$. Then $\tau_\gamma \in V_{qm}$ since $V_\bullet$ is a graded linear series, and

$$\tau_\gamma u^{qm} = \prod_{j=1}^{q} (t^{\beta_j} + r_{\beta_j}) = t^\gamma + \sum_{\emptyset \neq J \subset \{1,\ldots,q\}} \prod_{j \in J} r_{\beta_j} \prod_{j \notin J} t^{\beta_j}.$$

For any non-empty $J \subset \{1, \ldots, q\}$, the corresponding term is contained in $\mathfrak{m}_x^{e_J}$, where

$$e_J := (s+1)|J| + \sum_{j \notin J} |\beta_j| \geq |J| + \sum_{j=1}^{q} |\beta_j| \geq |\gamma| + 1,$$

because $|\beta_j| \leq s$ for any $j \in J$. Thus

$$\tau_\gamma u^{qm} \equiv t^\gamma \mod \mathfrak{m}_x^{|\gamma|+1}$$

for any γ with $|\gamma| \leq qs$. Since the monomials t^γ with $|\gamma| = j$ span $\mathfrak{m}_x^j/\mathfrak{m}_x^{j+1}$, descending induction on j shows that the classes of $\tau_\gamma u^{qm}$ with $j \leq |\gamma| \leq qs$ span $\mathfrak{m}_x^j/\mathfrak{m}_x^{qs+1}$ for any $0 \leq j \leq qs$. For $j = 0$, the evaluation map $V_{qm} \to \mathcal{O}_{X,x}/\mathfrak{m}_x^{qs+1}$ is surjective. Therefore $s(V_{qm}, x) \geq qs$.

Let I be a positive integer. For any $m \geq 1$, V_{Im} is a linear subspace of $H^0(X, \mathcal{O}_X(m(ID)))$, and

$$\epsilon_{\mathrm{mov}}((V_{Im})_{m\geq 1}, x) = \limsup_{m \to +\infty} \frac{s(V_{Im}, x)}{m} = I \limsup_{m \to +\infty} \frac{s(V_{Im}, x)}{Im}.$$

Since $(s(V_{Im}, x)/(Im))_{m \geq 1}$ is a subsequence of $(s(V_n, x)/n)_{n \geq 1}$, we have

$$\epsilon_{\mathrm{mov}}((V_{Im})_{m\geq 1}, x) \leq I\epsilon_{\mathrm{mov}}(V_\bullet, x).$$

Taking $q = I$ in the first part, we have $s(V_{Im}, x) \geq Is(V_m, x)$ for any $m \geq 1$. Dividing by m and taking the limsup, we obtain $\epsilon_{\mathrm{mov}}((V_{Im})_{m\geq 1}, x) \geq I\epsilon_{\mathrm{mov}}(V_\bullet, x)$. □

Lemma 2.10. *Let $g : X \dashrightarrow X'$ be a birational map between normal projective varieties such that g^{-1} does not contract any divisor. Then the following hold.*

1. *$H^0(X, \mathcal{O}_X(-kK_X)) \subset H^0(X', \mathcal{O}_{X'}(-kK_{X'}))$ as subspaces of $\mathbb{C}(X) = \mathbb{C}(X')$ for any integer $k \geq 0$. In particular, $\operatorname{vol}(-K_X) \leq \operatorname{vol}(-K_{X'})$.*
2. *Let $U \subset X_{\mathrm{reg}}$ be an open subset such that $g|_U$ is an open immersion and $g(U) \subset X'_{\mathrm{reg}}$. Then $s(V, x) = s(V, g(x))$ for any $x \in U$, any integer $k \geq 1$, and any linear subspace $V \subset H^0(X, \mathcal{O}_X(-kK_X))$, where V is regarded as a subspace of $H^0(X', \mathcal{O}_{X'}(-kK_{X'}))$ by (1).*

Proof. (cf. [Zhu18, Lemma 3.2])

(1) Let $k \geq 0$ be an integer and $h \in H^0(X, \mathcal{O}_X(-kK_X))$ a non-zero element, so that $\operatorname{div}_X(h) - kK_X \geq 0$. Let P be a prime divisor on X'. Since g^{-1} does not contract any divisor, P is the birational transform of a prime divisor P_X on X, and $\operatorname{ord}_P = \operatorname{ord}_{P_X}$ as valuations of $\mathbb{C}(X) = \mathbb{C}(X')$. Thus

$$\operatorname{mult}_P \left(\operatorname{div}_{X'}(h) - kK_{X'} \right) = \operatorname{mult}_{P_X} \left(\operatorname{div}_X(h) - kK_X \right) \geq 0.$$

Since P is arbitrary, $\operatorname{div}_{X'}(h) - kK_{X'} \geq 0$, that is, $h \in H^0(X', \mathcal{O}_{X'}(-kK_{X'}))$. In particular, $h^0(X, \mathcal{O}_X(-kK_X)) \leq h^0(X', \mathcal{O}_{X'}(-kK_{X'}))$ for any $k \geq 1$, and $\dim X = \dim X'$. Hence $\operatorname{vol}(-K_X) \leq \operatorname{vol}(-K_{X'})$ by Definition 2.5.

(2) Let $x \in U$ and $k \geq 1$ an integer. Since $g|_U$ is an open immersion and $g(U) \subset X'_{\mathrm{reg}}$, we have

$$\mathcal{O}_{X',g(x)} = \mathcal{O}_{X,x}, \quad \mathfrak{m}_{g(x)} = \mathfrak{m}_x, \quad \mathcal{O}_{X'}(-kK_{X'})_{g(x)} = \mathcal{O}_X(-kK_X)_x$$

as subsets of $\mathbb{C}(X) = \mathbb{C}(X')$. Both evaluation maps send a section $h \in V$ to the class of the rational function h. Hence for any integer $s \geq 0$, under the identifications above the evaluation map

$$V \to \mathcal{O}_X(-kK_X) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s+1}$$

is equal to the evaluation map

$$V \to \mathcal{O}_{X'}(-kK_{X'}) \otimes \mathcal{O}_{X'}/\mathfrak{m}_{g(x)}^{s+1}.$$

Therefore $s(V, x) = s(V, g(x))$. □

The following theorem is [Zhu18, Theorem 1.10] combined with the sentence right after it, which states that X is then of Fano type. We use it in the proofs of Theorem 7.1 and Lemma 6.6.

Theorem 2.11 ([Zhu18, Theorem 1.10])**.** *Let X be a normal projective variety of dimension $d \geq 3$ such that $\epsilon_{\mathrm{mov}}(-K_X) = d$. Assume that there exists a birational map $g : X \dashrightarrow \mathbb{P}^d$ such that g^{-1} does not contract any divisor. Then X is of Fano type.*

2.3. **Valuations and normalized volumes.** In this subsection, we recall real valuations, their log discrepancies, and normalized volumes, following [BJ20, Sections 1.2, 1.4, 1.5, and 1.7]. The log canonical threshold used in Section 6 is computed by a real valuation v provided by Theorem 2.15(2), and v is not necessarily divisorial. We identify v by the equality case of normalized volumes (Lemma 4.3), and the uniqueness of minimizers of the normalized volume [XZ21, Theorem 1.1] is also a statement about real valuations. Thus we define log discrepancies, volumes, and normalized volumes on Val_X.

Definition 2.12. Let X be a normal variety.

(1) We denote by Val_X the set of valuations $v : \mathbb{C}(X)^* \to \mathbb{R}$ that are trivial on $\mathbb{C}^*$ and have a center on X. For $v \in \mathrm{Val}_X$, we denote by $\operatorname{center}_X v$ the closure of the center of v on X.

(2) For $v \in \mathrm{Val}_X$ and a non-zero ideal sheaf $\mathfrak{a} \subset \mathcal{O}_X$, we let

$$v(\mathfrak{a}) := \min\{v(h) \mid h \in \mathfrak{a} \cdot \mathcal{O}_{X,\xi}\},$$

where ξ is the generic point of $\operatorname{center}_X v$. For $t \in \mathbb{R}$, we let $\mathfrak{a}_t(v) := \{h \in \mathcal{O}_X \mid v(h) \geq t\}$. For a prime divisor E over X, we let $\operatorname{mult}_E \mathfrak{a} := v(\mathfrak{a})$ with $v = \operatorname{ord}_E$. For a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on X, we let $v(D) := v(\phi)/m$, where m is a positive integer such that mD is Cartier and ϕ is a local equation of mD at ξ.

(3) If K_X is $\mathbb{Q}$-Cartier, then we let $a(E, X) := a(E, X, 0)$ for any prime divisor E over X. If X is klt, then we denote by $a(\cdot, X) : \mathrm{Val}_X \to [0, +\infty]$ the extension of the log discrepancy in [BJ20, Section 1.5]. In particular, $a(c \operatorname{ord}_E, X) = ca(E, X)$ for any prime divisor E over X and any real number $c > 0$.

(4) For a smooth closed point $x \in X$, we denote by $\operatorname{ord}_x \in \mathrm{Val}_X$ the valuation defined by the exceptional divisor of the blow-up of X at x.

Definition 2.13. Let X be a normal variety of dimension d and $v \in \mathrm{Val}_X$ such that $x := \operatorname{center}_X v$ is a closed point. The *volume* of v is

$$\operatorname{vol}(v) := \lim_{t \to +\infty} \frac{d! \ell(\mathcal{O}_{X,x}/\mathfrak{a}_t(v))}{t^d},$$

where ℓ denotes the length (the limit exists by [BJ20, Section 1.4]). If X is klt, then the *normalized volume* of v, introduced in [Li18] (see also [XZ21, Definition 2.5]), is $\widehat{\operatorname{vol}}(v) := a(v, X)^d \operatorname{vol}(v)$ if

$a(v,X)<+\infty$, and $\widehat{\mathrm{vol}}(v):=+\infty$ otherwise. We let

$$\widehat{\mathrm{vol}}(X\ni x):=\inf\{\widehat{\mathrm{vol}}(w)\mid w\in \mathrm{Val}_X, \mathrm{center}_X w=x\}.$$

2.4. **Graded sequences of ideals and multiplier ideals.** In this subsection, we recall log canonical thresholds of graded sequences of ideals [BJ20, Sections 1.3 and 1.8], [Blu18, Section 2.9], and asymptotic multiplier ideals [BJ20, Sections 5.2–5.3].

Definition 2.14. Let X be a normal variety. A *graded sequence of ideals* on X is a sequence $\mathfrak{b}_\bullet=(\mathfrak{b}_m)_{m\geq 1}$ of ideal sheaves on X such that $\mathfrak{b}_m\cdot\mathfrak{b}_l\subset\mathfrak{b}_{m+l}$ for any $m,l\geq 1$ and $\mathfrak{b}_m\neq 0$ for some m. For $v\in\mathrm{Val}_X$, we let

$$v(\mathfrak{b}_\bullet):=\lim_{m\to+\infty}\frac{v(\mathfrak{b}_m)}{m}=\inf_m\frac{v(\mathfrak{b}_m)}{m},$$

where m runs through the positive integers such that $\mathfrak{b}_m\neq 0$ [BJ20, Section 1.3].

Assume that X is klt. For a non-zero ideal sheaf $\mathfrak{b}$ on X, the *log canonical threshold* of $\mathfrak{b}$ is $\mathrm{lct}(X;\mathfrak{b}):=\inf_E a(E,X)/\mathrm{mult}_E\mathfrak{b}$, where E runs through the prime divisors over X and $a(E,X)/0:=+\infty$. We let $\mathrm{lct}(X;0):=0$ and $\mathrm{lct}(X;\mathfrak{b}_\bullet):=\sup_{m\geq 1}m\,\mathrm{lct}(X;\mathfrak{b}_m)$. We say that $v\in\mathrm{Val}_X$ *computes* $\mathrm{lct}(X;\mathfrak{b}_\bullet)$ if $a(v,X)<+\infty$ and $\mathrm{lct}(X;\mathfrak{b}_\bullet)=a(v,X)/v(\mathfrak{b}_\bullet)$, where $a(v,X)/0:=+\infty$.

We will use the following results of Blum. Here [Blu18] assumes $\mathfrak{b}_m\neq 0$ for all m and excludes the trivial valuation from Val_X. The general case follows by applying it to $(\mathfrak{b}_{pm})_{m\geq 1}$ with $\mathfrak{b}_p\neq 0$, and Theorem 2.15(3) also holds with the trivial valuation in Val_X, since $0\leq v(\mathfrak{b}_\bullet)\leq v(\mathfrak{b}_p)/p$ for any $v\in\mathrm{Val}_X$ and the non-trivial valuations form an open subset of Val_X (for a non-trivial v and the ideal sheaf $\mathfrak{a}$ of $\mathrm{center}_X v\neq X$, the open subset $\{w\in\mathrm{Val}_X\mid w(\mathfrak{a})>0\}$ contains v but not the trivial valuation).

Theorem 2.15 ([Blu18, Sections 2.1 and 2.9, Theorem B.1, and Proposition B.2])**.** *Let X be a normal projective klt variety and $\mathfrak{b}_\bullet$ a graded sequence of ideals on X. Then the following hold.*

1. *$\mathrm{lct}(X;\mathfrak{b}_\bullet)=\inf_v a(v,X)/v(\mathfrak{b}_\bullet)$, where v runs through the non-trivial valuations in Val_X and $a(v,X)/0:=+\infty$.*
2. *If $\mathrm{lct}(X;\mathfrak{b}_\bullet)<+\infty$, then there exists $v\in\mathrm{Val}_X$ computing $\mathrm{lct}(X;\mathfrak{b}_\bullet)$.*
3. *Equip Val_X with the weakest topology such that $v\to v(\mathfrak{a})$ is continuous for any non-zero ideal sheaf $\mathfrak{a}$ on X. Then the function $v\to v(\mathfrak{b}_\bullet)$ is continuous on $\{v\in\mathrm{Val}_X\mid a(v,X)<+\infty\}$.*

Definition 2.16. Let X be a normal projective klt variety.

(1) For a non-zero ideal sheaf $\mathfrak{a}$ on X and a rational number $c>0$, the *multiplier ideal* of $\mathfrak{a}$ with exponent c is

$$\mathcal{J}(X,c\cdot\mathfrak{a}):=\mu_*\mathcal{O}_W(\lceil K_W-\mu^*K_X-cF\rceil),$$

where $\mu:W\to X$ is a log resolution of $\mathfrak{a}$ and $\mathfrak{a}\cdot\mathcal{O}_W=\mathcal{O}_W(-F)$. We let $\mathcal{J}(X,c\cdot 0):=0$.

(2) For a graded sequence of ideals $\mathfrak{b}_\bullet$ on X and a rational number $c>0$, the *asymptotic multiplier ideal* $\mathcal{J}(X,c\cdot\mathfrak{b}_\bullet)$ is the unique maximal element of the set of ideals $\mathcal{J}(X,(c/p)\cdot\mathfrak{b}_p)$ with $p\geq 1$ [BJ20, Section 5.3].

For a Cartier divisor L on X and a linear subspace $V\subset H^0(X,\mathcal{O}_X(L))$, the *base ideal* of V is the image of the map $V\otimes\mathcal{O}_X(-L)\to\mathcal{O}_X$. If $V=0$, then the base ideal of V is 0.

Lemma 2.17 contains the inputs that Section 5 uses to compare sections of a graded linear series with lengths at points. By (1) and (2), the sections of the graded linear series are sections of a twist of the asymptotic multiplier ideal, and the higher cohomology of the twist vanishes. No vanishing theorem is available for the base ideal itself. Item (4) bounds the length of the quotient of $\mathcal{O}_X$ by the asymptotic multiplier ideal from below, and (3) controls the support of the quotient. We need (4) for real valuations, since Section 6 uses (4) for a valuation computing a

log canonical threshold, and the usual argument, which compares coefficients on a log resolution, only works for divisorial valuations.

Lemma 2.17. *Let X be a normal projective klt variety, L a big Cartier divisor on X, and $V_\bullet$ a graded linear series of L such that $V_m \neq 0$ for some m. For any $m \geq 1$, let $\mathfrak{b}_m$ be the base ideal of V_m, and for any integer $k \geq 1$, let $\mathcal{J}_k := \mathcal{J}(X, k \cdot \mathfrak{b}_\bullet)$. Then the following hold for any positive integers k and m.*

(1) $\mathfrak{b}_k \subset \mathcal{J}_k$.
(2) If L' is a Cartier divisor on X such that $L' - K_X - kL$ is big and nef, then $H^i(X, \mathcal{O}_X(L') \otimes \mathcal{J}_k) = 0$ for any $i \geq 1$.
(3) $\mathfrak{b}_m^k \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}_k^m \cdot \mathcal{O}_{X_{\mathrm{reg}}}$.
(4) $\mathcal{J}_k \subset \mathfrak{a}_{kv(\mathfrak{b}_\bullet) - a(v,X)}(v)$ for any $v \in \mathrm{Val}_X$ such that $a(v, X) < +\infty$.

Proof. Since $V_m \cdot V_l \subset V_{m+l}$ for any $m, l \geq 1$, we have $\mathfrak{b}_m \cdot \mathfrak{b}_l \subset \mathfrak{b}_{m+l}$. Moreover, $\mathfrak{b}_m \neq 0$ whenever $V_m \neq 0$. Thus $\mathfrak{b}_\bullet$ is a graded sequence of ideals, and $\mathcal{J}_k$ is defined.

By [BJ20, Section 5.4], the multiplier ideal of the graded linear series $V_\bullet$ with exponent k is by definition $\mathcal{J}(X, k \cdot \mathfrak{b}_\bullet) = \mathcal{J}_k$. Thus (1) is [BJ20, Lemma 5.8(ii)], and (2) is [BJ20, Theorem 5.9(i)], a consequence of Nadel vanishing [Nad90], applied to the line bundle $\mathcal{O}_X(L')$.

(3) If $\mathfrak{b}_m = 0$, then there is nothing to prove, so we may assume that $\mathfrak{b}_m \neq 0$. Let Jac_X be the Jacobian ideal of X as in [BJ20, Theorem 5.5]. Since X_{reg} is smooth, $\mathrm{Jac}_X \cdot \mathcal{O}_{X_{\mathrm{reg}}} = \mathcal{O}_{X_{\mathrm{reg}}}$. Thus [BJ20, Theorem 5.5] implies that

$$\mathcal{J}(X, c \cdot (\mathfrak{a} \cdot \mathfrak{a}')) \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}(X, c \cdot \mathfrak{a}) \cdot \mathcal{J}(X, c \cdot \mathfrak{a}') \cdot \mathcal{O}_{X_{\mathrm{reg}}}$$

for any non-zero ideal sheaves $\mathfrak{a}, \mathfrak{a}'$ on X and any rational number $c > 0$. Applying the inclusion with $c = k/m$, $\mathfrak{a} = \mathfrak{b}_m$, and $\mathfrak{a}' = \mathfrak{b}_m^j$ for $j = 1, \ldots, m-1$, we obtain

$$\mathcal{J}(X, (k/m) \cdot \mathfrak{b}_m^m) \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}(X, (k/m) \cdot \mathfrak{b}_m)^m \cdot \mathcal{O}_{X_{\mathrm{reg}}}.$$

A log resolution of $\mathfrak{b}_m$ is a log resolution of $\mathfrak{b}_m^k$ and of $\mathfrak{b}_m^m$, so Definition 2.16(1) implies

$$\mathcal{J}(X, 1 \cdot \mathfrak{b}_m^k) = \mathcal{J}(X, k \cdot \mathfrak{b}_m) = \mathcal{J}(X, (k/m) \cdot \mathfrak{b}_m^m).$$

By [BJ20, Lemma 5.4(1)], $\mathfrak{b}_m^k \subset \mathcal{J}(X, 1 \cdot \mathfrak{b}_m^k)$. By Definition 2.16(2), $\mathcal{J}(X, (k/m) \cdot \mathfrak{b}_m) \subset \mathcal{J}_k$. Therefore

$$\mathfrak{b}_m^k \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}(X, k \cdot \mathfrak{b}_m) \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}(X, (k/m) \cdot \mathfrak{b}_m)^m \cdot \mathcal{O}_{X_{\mathrm{reg}}} \subset \mathcal{J}_k^m \cdot \mathcal{O}_{X_{\mathrm{reg}}}.$$

(4) Let $\lambda := v(\mathfrak{b}_\bullet)$. If $\lambda = 0$, then $\mathfrak{a}_{-a(v,X)}(v) = \mathcal{O}_X$ since $a(v, X) \geq 0$, and there is nothing to prove. Thus we may assume that $\lambda > 0$. By Definition 2.14, $v(\mathfrak{b}_p) \geq p\lambda$ whenever $\mathfrak{b}_p \neq 0$, so $\mathfrak{b}_p \subset \mathfrak{a}_{p\lambda}(v)$ for any $p \geq 1$. Since $\lambda > 0$, v is non-trivial. We let $\mathfrak{a}_\bullet(v) := (\mathfrak{a}_m(v))_{m \geq 1}$. Since $v(gh) = v(g) + v(h)$, we have $\mathfrak{a}_m(v) \cdot \mathfrak{a}_l(v) \subset \mathfrak{a}_{m+l}(v)$ for any $m, l \geq 1$. Moreover, for any $m \geq 1$, pick p such that $\mathfrak{b}_p \neq 0$ and an integer $N \geq 1$ such that $Np\lambda \geq m$. Then $0 \neq \mathfrak{b}_p^N \subset \mathfrak{a}_{Np\lambda}(v) \subset \mathfrak{a}_m(v)$. Thus $\mathfrak{a}_\bullet(v)$ is a graded sequence of ideals in the sense of Definition 2.14.

Let λ' be a rational number such that $0 < \lambda' < \lambda$. By [BJ20, Section 5.3], there exists a positive integer p such that $p\lambda'$ is an integer and $\mathcal{J}_k = \mathcal{J}(X, (k/p) \cdot \mathfrak{b}_p)$. Since $\mathfrak{b}_p \subset \mathfrak{a}_{p\lambda}(v) \subset \mathfrak{a}_{p\lambda'}(v)$, [BJ20, Lemma 5.4(2)] implies

$$\mathcal{J}_k = \mathcal{J}(X, (k/p) \cdot \mathfrak{b}_p) \subset \mathcal{J}(X, (k/p) \cdot \mathfrak{a}_{p\lambda'}(v)).$$

Since $(k\lambda')/(p\lambda') = k/p$, the ideal $\mathcal{J}(X, (k/p) \cdot \mathfrak{a}_{p\lambda'}(v))$ is the element of index $p\lambda'$ in the set of Definition 2.16(2) for $\mathfrak{a}_\bullet(v)$ and $c = k\lambda'$. Thus

$$\mathcal{J}(X, (k/p) \cdot \mathfrak{a}_{p\lambda'}(v)) \subset \mathcal{J}(X, k\lambda' \cdot \mathfrak{a}_\bullet(v)) \subset \mathfrak{a}_{k\lambda' - a(v,X)}(v),$$

where the last inclusion is [BJ20, Proposition 5.7]. Hence $v(h) \geq k\lambda' - a(v, X)$ for any local section h of $\mathcal{J}_k$ at the generic point of $\mathrm{center}_X v$. Letting $\lambda' \to \lambda$, we obtain $\mathcal{J}_k \subset \mathfrak{a}_{k\lambda - a(v,X)}(v)$. □

Lemma 2.18. *Let X be a normal projective klt variety, L a Cartier divisor on X, $V \subset H^0(X, \mathcal{O}_X(L))$ a non-zero linear subspace with base ideal $\mathfrak{b}$, and t a real number such that $0 < t < 1$ and $t < \mathrm{lct}(X; \mathfrak{b})$. Then (X, tG) is klt for a general member G of $|V|$.*

Proof. Since $G \geq 0$, $a(E, X, tG) \geq a(E, X, t'G)$ for any prime divisor E over X and any $t \leq t'$. Choosing a rational number t' with $t < t' < \min\{1, \operatorname{lct}(X; \mathfrak{b})\}$, we may assume that t is rational.

Let $h : X' \to X$ be a log resolution of $(X, \mathfrak{b})$, write $h^*|V| = F + |M|$, where $\mathfrak{b} \cdot \mathcal{O}_{X'} = \mathcal{O}_{X'}(-F)$ and $|M|$ is base point free, and write $K_{X'} + B' := h^*K_X$. For any prime divisor E on X', $\operatorname{mult}_E(B' + tF) = 1 - a(E, X) + t \operatorname{mult}_E \mathfrak{b} < 1$ since $t < \operatorname{lct}(X; \mathfrak{b})$, so $(X', B' + tF)$ is sub-klt by [KM98, Corollary 2.31(3)]. For a general member G of $|V|$, $h^*G = F + M_G$ for a general member M_G of $|M|$, and $(X', B' + tF + tM_G)$ is sub-klt by [KM98, Lemma 5.17(2)] and the linearity of discrepancies in the boundary, since $t < 1$. Since $h^*(K_X + tG) = K_{X'} + B' + tF + tM_G$, (X, tG) is klt by [KM98, Lemma 2.30]. □

When X is smooth, Lemma 2.18 also follows from [Laz04b, Proposition 9.2.26].

3. Reduction to terminal varieties

In this section, we prove Lemma 3.1 and Proposition 3.2, which reduce Theorem 1.2 to $\mathbb{Q}$-factorial terminal varieties Y of dimension $d \geq 3$ with $\epsilon_{\mathrm{mov}}(-K_Y) = d$ and $\operatorname{vol}(-K_Y) = d^d$.

In Theorem 1.2, X need not be klt, and K_X need not be $\mathbb{Q}$-Cartier, so [Zhu18, Proposition 1.9] cannot be used for X directly. Lemma 3.1 replaces X with a $\mathbb{Q}$-factorial terminal variety Y such that X and Y have equal spaces of anti-canonical sections in each degree (Lemma 3.1(1)), and such that Fano type descends from Y to X.

Lemma 3.1. *Let X be a normal projective variety. Then there exists a projective birational morphism $f : Y \to X$ such that Y is $\mathbb{Q}$-factorial and terminal and K_Y is nef over X. Moreover, for any such f, the following hold.*

(1) $H^0(X, \mathcal{O}_X(-kK_X)) = H^0(Y, \mathcal{O}_Y(-kK_Y))$ as subspaces of $\mathbb{C}(X) = \mathbb{C}(Y)$ for any integer $k \geq 0$.

(2) $\epsilon_{\mathrm{mov}}(-K_X) = \epsilon_{\mathrm{mov}}(-K_Y)$ and $\operatorname{vol}(-K_X) = \operatorname{vol}(-K_Y)$.

(3) If Y is of Fano type, then X is of Fano type.

Proof. The existence of f follows from [BCHM10, Corollary 1.4.2] (cf. the paragraph after [Zhu18, Definition 2.4], where the existence of terminal modifications is derived from [BCHM10]). To construct f, we run a K_W-MMP over X with scaling, starting from a log resolution W of X, which terminates by [BCHM10, Corollary 1.4.2]. Since $W \to X$ is birational, the MMP ends with a higher model Y of X on which K_Y is nef over X, and Y is $\mathbb{Q}$-factorial and terminal since W is smooth and each step of the MMP preserves both properties [KM98, Propositions 3.36(1), 3.37(1) and Corollaries 3.42, 3.43(3)]. Now let $f : Y \to X$ be any projective birational morphism such that Y is $\mathbb{Q}$-factorial and terminal and K_Y is nef over X. We prove (1)–(3). Since f is a projective birational morphism, f^{-1} does not contract any divisor by the paragraph before Theorem 2.3.

(1) We follow [Zhu18, Lemma 3.4]. For $k = 0$, both sides are equal to $\mathbb{C}$. Let $k \geq 1$ be an integer. By Lemma 2.10(1) applied to f, we have

$$H^0(Y, \mathcal{O}_Y(-kK_Y)) \subset H^0(X, \mathcal{O}_X(-kK_X)).$$

For the reverse inclusion, let $u \in H^0(X, \mathcal{O}_X(-kK_X))$ be a non-zero element, and let

$$D := \operatorname{div}_X(u) - kK_X \geq 0.$$

Then $K_X + \frac{1}{k}D = \frac{1}{k}\operatorname{div}_X(u)$ is $\mathbb{Q}$-Cartier, so $(X, \frac{1}{k}D)$ is a pair. By [Zhu18, Lemma 2.5], $a(E, X, \frac{1}{k}D) \leq 1$ for any f-exceptional prime divisor E on Y. Thus we may write

$$f^*\Big(K_X + \frac{1}{k}D\Big) = K_Y + \frac{1}{k}f_*^{-1}D + \frac{1}{k}F_u,$$

where F_u is f-exceptional and $\operatorname{mult}_E F_u = k\big(1 - a(E, X, \frac{1}{k}D)\big) \geq 0$ for any f-exceptional prime divisor E on Y. In particular, $F_u \geq 0$. Since $f^* \operatorname{div}_X(u) = \operatorname{div}_Y(u)$, we have

$$\operatorname{div}_Y(u) = kf^*\Big(K_X + \frac{1}{k}D\Big) = kK_Y + f_*^{-1}D + F_u.$$

Moreover, $F_u = \mathrm{div}_Y(u) - kK_Y - f_*^{-1}D$ is a divisor, since $\mathrm{div}_Y(u)$, kK_Y, and $f_*^{-1}D$ are divisors. Thus $\mathrm{div}_Y(u) - kK_Y = f_*^{-1}D + F_u \geq 0$, that is, $u \in H^0(Y, \mathcal{O}_Y(-kK_Y))$. This implies (1).

(2) Let $U \subset Y$ be a dense open subset such that $f|_U$ is an open immersion and $f(U) \subset X_{\mathrm{reg}}$. Then $U \subset Y_{\mathrm{reg}}$. By (1) and Lemma 2.10(2) applied to f and $V = H^0(Y, \mathcal{O}_Y(-kK_Y))$, we have

$$s\big(H^0(Y, \mathcal{O}_Y(-kK_Y)), y\big) = s\big(H^0(X, \mathcal{O}_X(-kK_X)), f(y)\big)$$

for any $y \in U$ and any integer $k \geq 1$. Hence $\epsilon_{\mathrm{mov}}(-K_Y, y) = \epsilon_{\mathrm{mov}}(-K_X, f(y))$ for any $y \in U$. Let $y \in U$ be a very general point. Since $f|_U$ is an open immersion, $f(y)$ is a very general point of X. By Lemma 2.8,

$$\epsilon_{\mathrm{mov}}(-K_Y) = \epsilon_{\mathrm{mov}}(-K_Y, y) = \epsilon_{\mathrm{mov}}(-K_X, f(y)) = \epsilon_{\mathrm{mov}}(-K_X).$$

By (1), we have

$$h^0(X, \mathcal{O}_X(-kK_X)) = h^0(Y, \mathcal{O}_Y(-kK_Y))$$

for any integer $k \geq 1$, hence $\mathrm{vol}(-K_X) = \mathrm{vol}(-K_Y)$ by Definition 2.5.

(3) Since f^{-1} does not contract any divisor, (3) follows from Theorem 2.3 applied to f. □

Proposition 3.2. *Let X be a normal projective variety of dimension $d \geq 1$. Assume that $\epsilon_{\mathrm{mov}}(-K_X) \geq d$. Then $\mathrm{vol}(-K_X) \geq \epsilon_{\mathrm{mov}}(-K_X)^d$, and the following hold.*

(1) If $\epsilon_{\mathrm{mov}}(-K_X) > d$, then $X \cong \mathbb{P}^d$. In particular, X is of Fano type.
(2) If $\epsilon_{\mathrm{mov}}(-K_X) = d$ and $\mathrm{vol}(-K_X) > d^d$, then X is of Fano type.
(3) If $d \leq 2$, then X is of Fano type.

Proof. We first prove $\mathrm{vol}(-K_X) \geq \epsilon_{\mathrm{mov}}(-K_X)^d$ together with (1).

Suppose that $\epsilon_{\mathrm{mov}}(-K_X) > d$. By [Zhu18, Theorem 1.7], $X \cong \mathbb{P}^d$. Let $k \geq 1$ be an integer. Then

$$H^0(X, \mathcal{O}_X(-kK_X)) = H^0(\mathbb{P}^d, \mathcal{O}_{\mathbb{P}^d}(k(d+1))).$$

Let $x \in \mathbb{P}^d$ be a point. In affine coordinates centered at x, the elements of $H^0(\mathbb{P}^d, \mathcal{O}_{\mathbb{P}^d}(k(d+1)))$ restrict to the polynomials of degree at most $k(d+1)$, so the evaluation map

$$H^0(\mathbb{P}^d, \mathcal{O}_{\mathbb{P}^d}(k(d+1))) \to \mathcal{O}_{\mathbb{P}^d}(k(d+1)) \otimes \mathcal{O}_{\mathbb{P}^d}/\mathfrak{m}_x^{k(d+1)+1}$$

is surjective. Since

$$h^0(\mathbb{P}^d, \mathcal{O}_{\mathbb{P}^d}(k(d+1))) = \binom{d+k(d+1)}{d} < \binom{d+k(d+1)+1}{d} = \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^d,x}/\mathfrak{m}_x^{k(d+1)+2},$$

we have $s(H^0(X, \mathcal{O}_X(-kK_X)), x) = k(d+1)$. Thus $\epsilon_{\mathrm{mov}}(-K_X) = d+1$, and

$$\mathrm{vol}(-K_X) = \lim_{k\to+\infty} \frac{d!}{k^d}\binom{d+k(d+1)}{d} = (d+1)^d = \epsilon_{\mathrm{mov}}(-K_X)^d.$$

Moreover, $(\mathbb{P}^d, 0)$ is klt and $-K_{\mathbb{P}^d} = \mathcal{O}_{\mathbb{P}^d}(d+1)$ is ample, so X is of Fano type. This implies (1).

Suppose that $\epsilon_{\mathrm{mov}}(-K_X) = d$. Let $x \in X$ be a very general point. By Lemma 2.8, $x \in X_{\mathrm{reg}}$ and $\epsilon_{\mathrm{mov}}(-K_X, x) = d$. For any integer $k \geq 1$, let $s_k := s(H^0(X, \mathcal{O}_X(-kK_X)), x)$. Then there exists a sequence of positive integers k_j such that $\lim_{j\to+\infty} k_j = +\infty$ and $\lim_{j\to+\infty} s_{k_j}/k_j = d$. In particular, $s_{k_j} \geq 0$ for any $j \gg 0$. Since x is a smooth point and $\mathcal{O}_X(-k_jK_X)$ is invertible near x, for any $j \gg 0$, the evaluation map

$$H^0(X, \mathcal{O}_X(-k_jK_X)) \to \mathcal{O}_X(-k_jK_X) \otimes \mathcal{O}_X/\mathfrak{m}_x^{s_{k_j}+1}$$

is a surjection onto a $\mathbb{C}$-vector space of dimension $\binom{d+s_{k_j}}{d}$. Hence $h^0(X, \mathcal{O}_X(-k_jK_X)) \geq \binom{d+s_{k_j}}{d}$ for any $j \gg 0$. Since $d!\binom{d+s}{d} = (s+1)(s+2)\cdots(s+d)$ for any integer $s \geq 0$, we have

$$\mathrm{vol}(-K_X) \geq \limsup_{j\to+\infty} \frac{d!}{k_j^d}\binom{d+s_{k_j}}{d} = \lim_{j\to+\infty} \prod_{i=1}^{d} \frac{s_{k_j}+i}{k_j} = d^d = \epsilon_{\mathrm{mov}}(-K_X)^d.$$

Thus $\mathrm{vol}(-K_X) \geq \epsilon_{\mathrm{mov}}(-K_X)^d$ in both cases.

(2) This follows from [Zhu18, Corollary 4.1].

(3) Suppose that $d = 2$. If $\epsilon_{\mathrm{mov}}(-K_X) > 2$, then X is of Fano type by (1). If $\epsilon_{\mathrm{mov}}(-K_X) = 2$, then X is of Fano type by [Zhu18, Theorem 1.8(2)].

Suppose that $d = 1$. Since a normal curve is smooth [Har77, Theorem I.6.2A and Theorem I.5.1], X is a smooth projective curve. Let g be the genus of X, and suppose that $g \geq 1$. For any integer $k \geq 1$, we have

$$\deg(-kK_X) = k(2-2g) \leq 0,$$

so $h^0(X, \mathcal{O}_X(-kK_X)) \leq 1$ by [Har77, Lemma IV.1.2]. Since $\dim_{\mathbb{C}} \mathcal{O}_{X,x}/\mathfrak{m}_x^2 = 2$ for any $x \in X$, we have $s(H^0(X, \mathcal{O}_X(-kK_X)), x) \leq 0$ for any $x \in X$ and any integer $k \geq 1$. Thus $\epsilon_{\mathrm{mov}}(-K_X) \leq 0$, which contradicts $\epsilon_{\mathrm{mov}}(-K_X) \geq 1$. Therefore, $g = 0$, and $X \cong \mathbb{P}^1$ by [Har77, Example IV.1.3.5]. Since $(\mathbb{P}^1, 0)$ is klt and $-K_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(2)$ is ample, X is of Fano type. □

4. Weighted hypersurfaces

In this section, we study the weighted hypersurfaces of degree $r+1$ in $\mathbb{P}(1^{d+1}, r)$ which form the class of Theorem 7.2(1), and we prove Proposition 4.8 on the centers of valuations in Subsection 4.2. In the proof of Theorem 1.2, the weighted hypersurfaces form the only class that needs a new argument, which is given in this section, Section 5, and Section 6.

4.1. **Geometry of the weighted hypersurfaces.** In this subsection, we fix the notation for the weighted hypersurfaces and study their charts, singular points, canonical divisors, and normalized volumes. We refer to [Dol82] for weighted projective spaces and the sheaves $\mathcal{O}(m)$ on them.

Set-up 4.1. Let d, r, f, Z, H, o, π, and Γ be as follows.

(1) $d \geq 3$ and $r \geq 1$ are integers, and $f \in \mathbb{C}[x_1, \dots, x_d]$ is a non-zero homogeneous polynomial of degree $r+1$.
(2) $Z := (x_0 x_{d+1} = f(x_1, \dots, x_d)) \subset \mathbb{P}(1^{d+1}, r)$, where $x_0, \dots, x_d$ are homogeneous coordinates of weight 1 and x_{d+1} is a homogeneous coordinate of weight r, and Z is normal.
(3) H is a Weil divisor on Z such that $\mathcal{O}_Z(H) \cong \mathcal{O}_Z(1)$, and $o := [0 : \cdots : 0 : 1] \in Z$.
(4) $\pi : Z \dashrightarrow \mathbb{P}^d$ is the birational map defined by $x_0, \dots, x_d$.
(5) Γ is the prime divisor over Z such that ord_Γ is the valuation of $\mathbb{C}(Z) = \mathbb{C}(\mathbb{P}^d)$ defined by the hyperplane $(x_0 = 0) \subset \mathbb{P}^d$.

The prime divisor Γ is the prime divisor over Z that we look for in Section 6. Lemma 4.3 computes the center, the log discrepancy, and the volume of ord_Γ, and Lemma 4.4 uses Γ to pass from birational maps to Z to birational maps to $\mathbb{P}^d$.

Lemma 4.2. *Notation and conditions as in Set-up 4.1. Then the following hold.*

(1) The open subsets $Z \cap (x_i \neq 0)$, $0 \leq i \leq d$, cover $Z \backslash \{o\}$. The sheaf $\mathcal{O}_Z(1)$ is invertible on $Z \backslash \{o\}$, and $\mathcal{O}_Z(r)$ is an ample invertible sheaf on Z.
(2) A closed point $z \in Z \backslash \{o\}$ is a singular point of Z if and only if $z = [0 : u_1 : \cdots : u_d : 0]$ for some $u = (u_1, \dots, u_d) \neq 0$ such that $\frac{\partial f}{\partial x_i}(u) = 0$ for $1 \leq i \leq d$. Moreover, $f(u) = 0$ for any such u.
(3) The open subset $Z \cap (x_{d+1} \neq 0)$ is isomorphic to $\mathbb{A}^d/\mu_r$, where μ_r acts on $\mathbb{A}^d$ by scalar multiplication, and o is the image of the origin. In particular, $(Z \cap (x_{d+1} \neq 0)) \backslash \{o\} \subset Z_{\mathrm{reg}}$, and $o \in Z_{\mathrm{reg}}$ if $r = 1$.
(4) $K_Z \sim -dH$ and $(H^d) = (r+1)/r$. In particular, if Z is klt, then Z is a $\mathbb{Q}$-Fano variety and $\mathrm{vol}(-K_Z) = d^d(r+1)/r$.

Proof. Let $S := \mathbb{C}[x_0, \dots, x_{d+1}]$, graded so that $x_0, \dots, x_d$ have degree 1 and x_{d+1} has degree r, let $F := x_0 x_{d+1} - f(x_1, \dots, x_d) \in S_{r+1}$, and let $\mathbb{P} := \mathbb{P}(1^{d+1}, r) = \mathrm{Proj}\, S$.

(1) If $x_0, \dots, x_d$ vanish at a point $z \in Z$, then $z = o$. Thus the open subsets $Z \cap (x_i \neq 0)$, $0 \leq i \leq d$, cover $Z \backslash \{o\}$. For $0 \leq i \leq d$, x_i has degree 1, so $\mathcal{O}_{\mathbb{P}}(m)$ is generated by x_i^m on $(x_i \neq 0)$

for any $m \in \mathbb{Z}$. Hence $\mathcal{O}_{\mathbb{P}}(1)$ is invertible on $\mathbb{P}\backslash\{o\}$, and $\mathcal{O}_{\mathbb{P}}(m) \cong \mathcal{O}_{\mathbb{P}}(1)^{\otimes m}$ on $\mathbb{P}\backslash\{o\}$. Moreover, $\mathcal{O}_{\mathbb{P}}(r)$ is generated by x_{d+1} on $(x_{d+1} \neq 0)$, so $\mathcal{O}_{\mathbb{P}}(r)$ is invertible. It is ample, since a monomial of degree kr in S is a product of k monomials of degree r, so $\bigoplus_{k\geq 0} S_{kr}$ is generated by S_r, and $\mathcal{O}_{\mathbb{P}}(r)$ is the very ample sheaf $\mathcal{O}(1)$ on $\operatorname{Proj} \bigoplus_{k\geq 0} S_{kr} \cong \mathbb{P}$. Equivalently, $\mathbb{P}$ is the projective cone over the Veronese variety $v_r(\mathbb{P}^d)$ [Dol82, 1.2.3]. Restricting to Z, we obtain (1).

(2) On the chart $(x_0 \neq 0) \cong \mathbb{A}^{d+1}$ with coordinates x_j/x_0, $1 \leq j \leq d$, and x_{d+1}/x_0^r, the variety Z is the graph of the function $f(x_1/x_0, \ldots, x_d/x_0)$, so $Z \cap (x_0 \neq 0)$ is smooth. Let $z \in Z\backslash\{o\}$ be a closed point such that $x_0(z) = 0$. By (1), $x_i(z) \neq 0$ for some $1 \leq i \leq d$, and we may write $z = [0 : u_1 : \cdots : u_d : u_{d+1}]$ with $u_i = 1$. On the chart $(x_i \neq 0) \cong \mathbb{A}^{d+1}$, Z is defined by

$$g := y_0 y_{d+1} - f(y_1, \ldots, y_{i-1}, 1, y_{i+1}, \ldots, y_d),$$

where $y_j = x_j/x_i$ for $j \neq i$, $0 \leq j \leq d$, and $y_{d+1} = x_{d+1}/x_i^r$. We have

$$\frac{\partial g}{\partial y_0} = y_{d+1}, \quad \frac{\partial g}{\partial y_{d+1}} = y_0, \quad \text{and} \quad \frac{\partial g}{\partial y_j} = -\frac{\partial f}{\partial x_j}(y_1, \ldots, 1, \ldots, y_d) \ \text{ for } j \neq i,\ 1 \leq j \leq d.$$

By the Jacobian criterion [Har77, Theorem I.5.1], z is a singular point of Z if and only if $u_{d+1} = 0$ and $\frac{\partial f}{\partial x_j}(u) = 0$ for any $j \neq i$. In this case, $f(u) = y_0(z)y_{d+1}(z) - g(z) = 0$, since $y_0(z) = 0$ and $z \in Z$. By the Euler formula,

$$(r+1)f(u) = \sum_{j=1}^{d} u_j \frac{\partial f}{\partial x_j}(u) = \frac{\partial f}{\partial x_i}(u)$$

whenever $\frac{\partial f}{\partial x_j}(u) = 0$ for any $j \neq i$. Thus z is a singular point of Z if and only if $z = [0 : u : 0]$ and $\frac{\partial f}{\partial x_j}(u) = 0$ for $1 \leq j \leq d$. Conversely, if $u \neq 0$ and $\frac{\partial f}{\partial x_j}(u) = 0$ for $1 \leq j \leq d$, then $(r+1)f(u) = \sum_j u_j \frac{\partial f}{\partial x_j}(u) = 0$ by the Euler formula, so $[0 : u : 0] \in Z$, and the Jacobian criterion above, applied on a chart $(x_i \neq 0)$ with $u_i \neq 0$, shows that $[0 : u : 0]$ is a singular point of Z. This proves (2).

(3) The degree-zero part of $S[x_{d+1}^{-1}]$ is isomorphic to $\mathbb{C}[x_0, \ldots, x_d]^{\mu_r}$ by setting $x_{d+1} = 1$, where μ_r acts on $\mathbb{A}^{d+1}$ with coordinates $x_0, \ldots, x_d$ by scalar multiplication. Hence $(x_{d+1} \neq 0) \subset \mathbb{P}$ is isomorphic to $\mathbb{A}^{d+1}/\mu_r$, and $Z \cap (x_{d+1} \neq 0)$ is isomorphic to $(x_0 = f(x_1, \ldots, x_d))/\mu_r$, since taking μ_r-invariants is exact. Since $f(\zeta x) = \zeta^{r+1} f(x) = \zeta f(x)$ for any $\zeta \in \mu_r$, the projection $(x_0, \ldots, x_d) \to (x_1, \ldots, x_d)$ is a μ_r-equivariant isomorphism from $(x_0 = f(x_1, \ldots, x_d)) \subset \mathbb{A}^{d+1}$ onto $\mathbb{A}^d$, with inverse $(x_1, \ldots, x_d) \to (f(x_1, \ldots, x_d), x_1, \ldots, x_d)$. Thus $Z \cap (x_{d+1} \neq 0) \cong \mathbb{A}^d/\mu_r$, and o is the image of the origin. Since μ_r acts freely on $\mathbb{A}^d\backslash\{0\}$, the quotient map is étale over $(Z \cap (x_{d+1} \neq 0))\backslash\{o\}$, so $(Z \cap (x_{d+1} \neq 0))\backslash\{o\} \subset Z_{\mathrm{reg}}$. If $r = 1$, then $\mathbb{A}^d/\mu_1 = \mathbb{A}^d$, so $o \in Z_{\mathrm{reg}}$.

(4) The charts $(x_i \neq 0)$, $0 \leq i \leq d$, of $\mathbb{P}$ are isomorphic to $\mathbb{A}^{d+1}$ and cover $\mathbb{P}\backslash\{o\}$, so $\mathbb{P}\backslash\{o\} \subset \mathbb{P}_{\mathrm{reg}}$. The weighted projective space $\mathbb{P}$ is quasi-smooth since the affine cone over $\mathbb{P}$ is $\mathbb{A}^{d+2}$. By [Dol82, Theorem 3.3.4] applied to $\mathbb{P}$,

$$\omega_{\mathbb{P}}|_{\mathbb{P}_{\mathrm{reg}}} \cong \mathcal{O}_{\mathbb{P}}(-(d+1+r))|_{\mathbb{P}_{\mathrm{reg}}}.$$

Let $U := Z_{\mathrm{reg}}\backslash\{o\}$ and $W := \mathbb{P}\backslash(\{o\} \cup \operatorname{Sing} Z)$. Then U is a smooth closed subvariety of codimension one of the smooth variety W, defined by the section F of the invertible sheaf $\mathcal{O}_{\mathbb{P}}(r+1)|_W$. By adjunction [Har77, Proposition II.8.20],

$$\omega_U \cong \omega_W \otimes \mathcal{O}_{\mathbb{P}}(r+1)|_U \cong \mathcal{O}_{\mathbb{P}}(-d)|_U \cong \mathcal{O}_Z(-dH)|_U,$$

where the last isomorphism holds because $\mathcal{O}_Z(H) \cong \mathcal{O}_Z(1)$ is invertible on $Z\backslash\{o\}$ by (1). Since Z is normal and $d \geq 2$, $Z\backslash U = \operatorname{Sing} Z \cup \{o\}$ has codimension at least 2 in Z. Thus $K_Z \sim -dH$ by [Har77, Proposition II.6.5(b)].

By (1), there exists a Cartier divisor H_r on Z such that $\mathcal{O}_Z(H_r) \cong \mathcal{O}_Z(r)$. Then $H_r|_{Z\backslash\{o\}} \sim rH|_{Z\backslash\{o\}}$, so $rH \sim H_r$ by [Har77, Proposition II.6.5(b)]. In particular, rH is an ample Cartier

divisor and $\mathcal{O}_Z(krH) \cong \mathcal{O}_Z(kr)$ for any integer $k \geq 1$. Since F is a nonzerodivisor of degree $r+1$ in S, we have an exact sequence of graded S-modules

$$0 \to S(kr-r-1) \xrightarrow{\cdot F} S(kr) \to (S/(F))(kr) \to 0,$$

which induces an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}}(kr-r-1) \to \mathcal{O}_{\mathbb{P}}(kr) \to \mathcal{O}_Z(kr) \to 0.$$

By [Dol82, Theorem 1.4.1], $H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m)) = S_m$ for any $m \in \mathbb{Z}$ and $H^1(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(kr-r-1)) = 0$, so $h^0(Z, \mathcal{O}_Z(krH)) = \dim(S/(F))_{kr}$. The Hilbert series of $S/(F)$ is

$$\sum_{m \geq 0} \dim(S/(F))_m t^m = \frac{1-t^{r+1}}{(1-t)^{d+1}(1-t^r)}.$$

The pole at $t=1$ has order $d+1$ and leading term $\frac{r+1}{r}(1-t)^{-(d+1)}$, and the other poles are at the non-trivial r-th roots of unity and are simple. Hence

$$\dim(S/(F))_m = \frac{r+1}{r} \cdot \frac{m^d}{d!} + O(m^{d-1}), \quad \text{so} \quad h^0(Z, \mathcal{O}_Z(krH)) = \frac{(r+1)r^{d-1}}{d!}k^d + O(k^{d-1}).$$

By [Laz04a, Corollary 1.4.41] applied to the ample Cartier divisor rH,

$$h^0(Z, \mathcal{O}_Z(krH)) = \frac{((rH)^d)}{d!}k^d + O(k^{d-1}).$$

Thus $((rH)^d) = (r+1)r^{d-1}$, and

$$(H^d) = \frac{((rH)^d)}{r^d} = \frac{r+1}{r}.$$

Assume that Z is klt. Since $-rK_Z \sim drH$ is an ample Cartier divisor, Z is a $\mathbb{Q}$-Fano variety. By Theorem 2.6 and [Laz04a, Corollary 1.4.41], $\mathrm{vol}(-rK_Z) = ((drH)^d) = d^d r^{d-1}(r+1)$, and

$$\mathrm{vol}(-K_Z) = \frac{\mathrm{vol}(-rK_Z)}{r^d} = \frac{d^d(r+1)}{r}$$

by Theorem 2.6. □

The following lemma provides the lower bounds for normalized volumes used in Section 6, where Z is klt: at a smooth point q of Z, normalized volumes of valuations centered at q are at least d^d, with equality only for positive multiples of ord_q. At o, normalized volumes of valuations centered at o are at least d^d/r, with equality only for positive multiples of ord_Γ. The argument of Section 6 closes because $d^d/r = \mathrm{vol}(-K_Z) - d^d$ when Z is klt (Lemma 4.2(4)).

Lemma 4.3. *Notation and conditions as in Set-up 4.1. Then Γ is the exceptional divisor of the blow-up of Z at o, $\mathrm{center}_Z(\mathrm{ord}_\Gamma) = o$, $a(\Gamma, Z) = d/r$, and $\mathrm{vol}(\mathrm{ord}_\Gamma) = r^{d-1}$. Moreover, if Z is klt, then for any $v \in \mathrm{Val}_Z$ such that $a(v,Z) < +\infty$ and $q := \mathrm{center}_Z v$ is a closed point, the following hold.*

(1) If $q \in Z_{\mathrm{reg}}$, then $\widehat{\mathrm{vol}}(v) \geq d^d$, and equality holds if and only if $v = c\,\mathrm{ord}_q$ for some real number $c > 0$.

(2) If $q = o$, then $\widehat{\mathrm{vol}}(v) \geq d^d/r$, and equality holds if and only if $v = c\,\mathrm{ord}_\Gamma$ for some real number $c > 0$.

Proof. Let $Z^\circ := Z \cap (x_{d+1} \neq 0)$ and let $\rho : \mathbb{A}^d \to Z^\circ \cong \mathbb{A}^d/\mu_r$ be the quotient map of Lemma 4.2(3), induced by setting $x_{d+1} = 1$, where $x_1, \ldots, x_d$ are the coordinates of $\mathbb{A}^d$. Then $\rho^*(x_j/x_1) = x_j/x_1$ for $2 \leq j \leq d$ and $\rho^*(x_0/x_1) = f(x_1, \ldots, x_d)/x_1$, and ρ^* induces an isomorphism from $\mathbb{C}(Z)$ onto $\mathbb{C}(\mathbb{A}^d)^{\mu_r}$. Let ord_0 be the valuation of the exceptional divisor of the blow-up of $\mathbb{A}^d$ at the origin.

We first show that

$$\mathrm{ord}_0(\rho^* h) = r\,\mathrm{ord}_\Gamma(h) \quad \text{for any } h \in \mathbb{C}(Z)^*. \tag{4.1}$$

Let $y_j := x_j/x_1$ for $2 \le j \le d$. Then $\mathbb{C}(\mathbb{A}^d) = \mathbb{C}(y_2, \dots, y_d)(x_1)$, and ord_0 is the x_1-adic valuation of $\mathbb{C}(y_2, \dots, y_d)(x_1)$ over $\mathbb{C}(y_2, \dots, y_d)$. On the other hand, $\mathbb{C}(Z) = \mathbb{C}(\mathbb{P}^d) = \mathbb{C}(y_2, \dots, y_d)(y_0)$ with $y_0 := x_0/x_1$, and ord_Γ is the y_0-adic valuation over $\mathbb{C}(y_2, \dots, y_d)$, since $(x_0 = 0)$ is defined by $y_0 = 0$ on the chart $(x_1 \ne 0) \subset \mathbb{P}^d$. We have

$$\rho^* y_0 = \frac{f(x_1, \dots, x_d)}{x_1} = x_1^r f(1, y_2, \dots, y_d),$$

where $0 \ne f(1, y_2, \dots, y_d) \in \mathbb{C}(y_2, \dots, y_d)$ because f is a non-zero homogeneous polynomial. For a non-zero polynomial $h = \sum_k c_k y_0^k$ with $c_k \in \mathbb{C}(y_2, \dots, y_d)$, we obtain $\rho^* h = \sum_k c_k f(1, y_2, \dots, y_d)^k x_1^{rk}$, so

$$\mathrm{ord}_0(\rho^* h) = r \min\{k \mid c_k \ne 0\} = r \, \mathrm{ord}_\Gamma(h).$$

Since both sides of (4.1) are additive with respect to products, (4.1) holds for quotients of such polynomials, that is, for any $h \in \mathbb{C}(Z)^*$.

The coordinate ring of Z° is $R := \mathbb{C}[x_1, \dots, x_d]^{\mu_r}$, which is spanned by the monomials whose degree is divisible by r. For a non-zero polynomial $h \in \mathbb{C}[x_1, \dots, x_d]$, $\mathrm{ord}_0(h)$ is the smallest degree of a monomial of h. By (4.1), $\mathrm{ord}_\Gamma \ge 0$ on R and $\mathrm{ord}_\Gamma > 0$ on the maximal ideal of o in R. Thus $\mathrm{center}_Z(\mathrm{ord}_\Gamma) = o$. Moreover, for any $t > 0$, the ideal $\mathfrak{a}_t(\mathrm{ord}_\Gamma) \cap R$ is spanned by the monomials of degree divisible by r and at least rt. Hence

$$\ell(\mathcal{O}_{Z,o}/\mathfrak{a}_t(\mathrm{ord}_\Gamma)) = \dim_{\mathbb{C}} R/(\mathfrak{a}_t(\mathrm{ord}_\Gamma) \cap R) = \sum_{0 \le j < t} \binom{rj + d - 1}{d - 1} = \frac{r^{d-1} t^d}{d!} + O(t^{d-1}),$$

where j runs through the integers, and $\mathrm{vol}(\mathrm{ord}_\Gamma) = r^{d-1}$.

By [Zhu18, Section 5, item (1)] (where the dimension is n and the weight is d), Γ is the exceptional divisor of the blow-up of Z at o and $a(\Gamma, Z) = d/r$. In particular, $a(\Gamma, Z)^d \, \mathrm{vol}(\mathrm{ord}_\Gamma) = (d/r)^d r^{d-1} = d^d/r$.

Assume that Z is klt, and let v and q be as in the statement.

(1) Assume that $q \in Z_{\mathrm{reg}}$. Fix $t > 0$. Since $\mathrm{center}_Z v = q$, we have $\mathfrak{a}_t(v)|_{Z \setminus \{q\}} = \mathcal{O}_{Z \setminus \{q\}}$, and $\mathfrak{a}_t(v)_q$ is primary to the maximal ideal of $\mathcal{O}_{Z,q}$. By Theorem 2.15(1) applied to the graded sequence of ideals $(\mathfrak{a}_t(v)^m)_{m \ge 1}$, we have

$$\mathrm{lct}(Z; \mathfrak{a}_t(v)) \le \frac{a(v, Z)}{v(\mathfrak{a}_t(v))} \le \frac{a(v, Z)}{t}.$$

Since $\mathfrak{a}_t(v)$ is non-trivial only at q, the local log canonical threshold of $\mathfrak{a}_t(v)$ at the point q of the smooth variety Z_{reg} is $\mathrm{lct}(Z; \mathfrak{a}_t(v))$. By [dFEM04, Theorem 1.1],

$$\ell(\mathcal{O}_{Z,q}/\mathfrak{a}_t(v)) > \frac{d^d}{d! \, \mathrm{lct}(Z; \mathfrak{a}_t(v))^d} \ge \frac{d^d t^d}{d! a(v, Z)^d}.$$

Multiplying by $d!/t^d$ and letting $t \to +\infty$, we obtain $\mathrm{vol}(v) \ge d^d/a(v, Z)^d$, that is, $\widehat{\mathrm{vol}}(v) \ge d^d$ (cf. [Li18, Proposition 5.2]). Since the blow-up of Z_{reg} at q has discrepancy $d - 1$ along the exceptional divisor, $a(\mathrm{ord}_q, Z) = d$. Since $\ell(\mathcal{O}_{Z,q}/\mathfrak{m}_q^k) = \binom{k+d-1}{d}$ for any integer $k \ge 0$, $\mathrm{vol}(\mathrm{ord}_q) = 1$. Thus $\widehat{\mathrm{vol}}(\mathrm{ord}_q) = d^d = \widehat{\mathrm{vol}}(Z \ni q)$. Since $\widehat{\mathrm{vol}}(cw) = \widehat{\mathrm{vol}}(w)$ for any $w \in \mathrm{Val}_Z$ and any real number $c > 0$, and since $q \in Z$ is a klt point, (1) follows from [XZ21, Theorem 1.1].

(2) Assume that $q = o$. Since μ_r acts freely on $\mathbb{A}^d \setminus \{0\}$, ρ is étale over $Z^\circ \setminus \{o\}$, so $K_{\mathbb{A}^d \setminus \{0\}} = \rho^* K_{Z^\circ \setminus \{o\}}$. Since $\mathrm{codim}_{\mathbb{A}^d}\{0\} = d \ge 2$, we have $K_{\mathbb{A}^d} = \rho^* K_{Z^\circ}$ by [Har77, Proposition II.6.5(b)]. The morphism $\rho : (\mathbb{A}^d \ni 0) \to (Z \ni o)$ is a finite Galois morphism of degree r between klt germs. We have $\widehat{\mathrm{vol}}(\mathbb{A}^d \ni 0) = d^d$ by [Li18, Proposition 5.2]. By [XZ21, Theorem 1.3],

$$\widehat{\mathrm{vol}}(Z \ni o) = \frac{\widehat{\mathrm{vol}}(\mathbb{A}^d \ni 0)}{r} = \frac{d^d}{r}.$$

Thus $\widehat{\mathrm{vol}}(v) \geq d^d/r$, and ord_Γ attains $\widehat{\mathrm{vol}}(Z \ni o)$ since $\widehat{\mathrm{vol}}(\mathrm{ord}_\Gamma) = a(\Gamma, Z)^d \,\mathrm{vol}(\mathrm{ord}_\Gamma) = d^d/r$. By [XZ21, Theorem 1.1], equality holds if and only if $v = c\,\mathrm{ord}_\Gamma$ for some real number $c > 0$. □

Lemma 4.4. *Notation and conditions as in Set-up 4.1. Let X be a normal projective variety and $g : X \dashrightarrow Z$ a birational map such that g^{-1} does not contract any divisor. Assume that $\mathrm{center}_X(\mathrm{ord}_\Gamma)$ is a prime divisor on X. Then $(\pi \circ g)^{-1}$ does not contract any divisor.*

Proof. Let P be a prime divisor on $\mathbb{P}^d$. We show that $\mathrm{center}_X(\mathrm{ord}_P)$ is a prime divisor on X. If $P = (x_0 = 0)$, then $\mathrm{ord}_P = \mathrm{ord}_\Gamma$, and the claim is our assumption. Assume that $P \neq (x_0 = 0)$. Then $P \cap (x_0 \neq 0) \neq \emptyset$. On the chart $(x_0 \neq 0)$ of $\mathbb{P}(1^{d+1}, r)$ with coordinates x_j/x_0, $1 \leq j \leq d$, and x_{d+1}/x_0^r, the map π is the projection $(x_1, \dots, x_d, x_{d+1}) \to (x_1, \dots, x_d)$, which restricts to an isomorphism from $Z \cap (x_0 \neq 0)$ onto $(x_0 \neq 0) \subset \mathbb{P}^d$, with inverse $(x_1, \dots, x_d) \to (x_1, \dots, x_d, f(x_1, \dots, x_d))$. Let P_Z be the closure of $\pi^{-1}(P \cap (x_0 \neq 0))$ in Z. Then P_Z is a prime divisor on Z and $\mathrm{ord}_P = \mathrm{ord}_{P_Z}$. Since g^{-1} does not contract any divisor, $\mathrm{center}_X(\mathrm{ord}_{P_Z})$ is a prime divisor on X. □

When $r = 1$, the point o is a smooth point of Z (Lemma 4.2(3)), and Section 6 constructs the valuation ord_q for some smooth point q, which need not be o. The following lemma moves q to o, so that the cases $r = 1$ and $r \geq 2$ are both treated with the prime divisor Γ of Set-up 4.1(5).

Lemma 4.5. *Notation and conditions as in Set-up 4.1. Assume that $r = 1$, and let $q \in Z_{\mathrm{reg}}$ be a closed point. Then there exist homogeneous coordinates $y_0, \dots, y_{d+1}$ on $\mathbb{P}^{d+1} = \mathbb{P}(1^{d+2})$ and a non-zero quadratic form $f' \in \mathbb{C}[y_1, \dots, y_d]$ such that, in the coordinates $y_0, \dots, y_{d+1}$, we have $q = [0 : \cdots : 0 : 1]$ and $Z = (y_0 y_{d+1} = f'(y_1, \dots, y_d))$. In particular, the conditions of Set-up 4.1 hold for $y_0, \dots, y_{d+1}$ and f', and the point o of Set-up 4.1 defined in the coordinates $y_0, \dots, y_{d+1}$ is q.*

Proof. Let $V := \mathbb{C}^{d+2}$ with coordinates $x_0, \dots, x_{d+1}$, let $Q := x_0 x_{d+1} - f(x_1, \dots, x_d)$, which is a quadratic form on V, and let B be the symmetric bilinear form on V such that $B(w, w) = Q(w)$ for any $w \in V$. The partial derivatives of Q at $w \in V$ are the coefficients of the linear form $2B(w, \cdot)$. By the Jacobian criterion [Har77, Theorem I.5.1] and the Euler formula, a closed point $[w] \in Z$ is a singular point of Z if and only if $B(w, \cdot) = 0$.

Write $q = [v]$. Then $Q(v) = 0$ and $B(v, \cdot) \neq 0$, so there exists $w \in V$ such that $B(v, w) = 1/2$. Let $w' := w - Q(w)v$. Then

$$Q(w') = Q(w) - 2Q(w)B(v, w) + Q(w)^2 Q(v) = 0 \quad \text{and} \quad B(v, w') = \frac{1}{2}.$$

Thus $Q(av + bw') = ab$ for any $a, b \in \mathbb{C}$. Let $V' := \{u \in V \mid B(u, v) = B(u, w') = 0\}$. Since B is non-degenerate on the span of v and w', we have $V = \mathbb{C}v \oplus \mathbb{C}w' \oplus V'$ and $\dim V' = d$. Let $e_1, \dots, e_d$ be a basis of V', and let $y_0, \dots, y_{d+1}$ be the linear forms on V such that

$$u = y_{d+1}(u)v + y_0(u)w' + \sum_{i=1}^{d} y_i(u)e_i$$

for any $u \in V$. Let $f'(y_1, \dots, y_d) := -Q(\sum_{i=1}^d y_i e_i)$. Since $B(v, e_i) = B(w', e_i) = 0$ for $1 \leq i \leq d$, we have

$$Q = y_0 y_{d+1} - f'(y_1, \dots, y_d).$$

Thus $Z = (y_0 y_{d+1} = f'(y_1, \dots, y_d))$, and $q = [v] = [0 : \cdots : 0 : 1]$ in the coordinates $y_0, \dots, y_{d+1}$. If $f' = 0$, then $Z = (y_0 y_{d+1} = 0)$ is reducible, which is not possible since Z is a normal variety. Hence f' is a non-zero quadratic form.

Since Z is normal, $d \geq 3$, $r = 1$, and f' is a non-zero homogeneous polynomial of degree $2 = r + 1$, the conditions of Set-up 4.1 hold for $y_0, \dots, y_{d+1}$ and f', and the point o defined in the coordinates $y_0, \dots, y_{d+1}$ is q. □

4.2. **Centers of valuations with large jet orders.** In this subsection, we show that if a sequence $V_\bullet$ of linear subspaces of $H^0(Z, \mathcal{O}_Z(dk))$ satisfies $\epsilon_{\mathrm{mov}}(V_\bullet, x) \geq d$ at a very general point $x \in Z$ and vanishes to order linear in k along a prime divisor E over Z, then $\mathrm{center}_Z(\mathrm{ord}_E)$ is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$ (Proposition 4.8). In Section 5, we use Proposition 4.8 to show that the cosupports of the asymptotic multiplier ideals there are isolated at such points, so that we can compute lengths at one point.

The curves of Lemma 4.7(2) passing through p_Z are contained in the divisor D_p of Construction 4.6. Thus they do not cover Z, and the center of a valuation need not be contained in one of them. Instead, we use three properties of D_p. First, D_p is an ample Cartier divisor, hence has non-empty intersection with any positive-dimensional closed subvariety of Z. Second, D_p does not contain o. Third, D_p contains any singular point of Z other than o.

Construction 4.6. Notation and conditions as in Set-up 4.1. For any $p = (p_1, \dots, p_d) \in \mathbb{C}^d$, we let $p_Z := [1 : p_1 : \cdots : p_d : f(p)] \in Z$. We let

$$G_p := x_{d+1} + \frac{f(x_1 - p_1 x_0, \dots, x_d - p_d x_0) - f(x_1, \dots, x_d)}{x_0},$$

which is a polynomial, since the numerator becomes zero after substituting $x_0 = 0$ and hence is divisible by x_0. We let $D_p := (G_p = 0) \subset Z$.

Since $x_0 x_{d+1} = f(x_1, \dots, x_d)$ on Z, we have $x_0 G_p = f(x_1 - p_1 x_0, \dots, x_d - p_d x_0)$ on Z. Thus, on the chart $Z \cap (x_0 \neq 0) \cong \mathbb{A}^d$ with coordinates $x_1/x_0, \dots, x_d/x_0$, the divisor D_p is defined by

$$f\Big(\frac{x_1}{x_0} - p_1, \dots, \frac{x_d}{x_0} - p_d\Big) = 0,$$

that is, $D_p \cap (x_0 \neq 0)$ is the cone $(f = 0)$ translated to have vertex p. Correspondingly, $\pi(D_p)$ is the cone in $\mathbb{P}^d$ with vertex $\pi(p_Z)$ over $(x_0 = f = 0)$, and the curves of Lemma 4.7(2) are the closures of the lines through p contained in the translated cone.

Lemma 4.7. *Notation and conditions as in Set-up 4.1 and Construction 4.6. Let $p \in \mathbb{C}^d$. Then the following hold.*

(1) G_p is a weighted homogeneous polynomial of degree r, and D_p is an ample Cartier divisor on Z such that $p_Z \in D_p$ and $o \notin D_p$.
(2) For any closed point $z \in D_p \backslash \{p_Z\}$, there exists a smooth rational curve $C \subset D_p$ such that $p_Z, z \in C$ and $(H \cdot C) = 1$.
(3) D_p contains any singular point of Z other than o.

Proof. By the definition of G_p, we have the identity of polynomials

$$x_0 G_p - f(x_1 - p_1 x_0, \dots, x_d - p_d x_0) = x_0 x_{d+1} - f(x_1, \dots, x_d), \tag{4.2}$$

and the right-hand side of (4.2) vanishes on Z. Moreover, the coefficient of x_0 in the polynomial

$$f(x_1 - p_1 x_0, \dots, x_d - p_d x_0) \in \mathbb{C}[x_1, \dots, x_d][x_0]$$

is $-\sum_{i=1}^d p_i \frac{\partial f}{\partial x_i}(x_1, \dots, x_d)$. Hence

$$G_p(0, x_1, \dots, x_{d+1}) = x_{d+1} - \sum_{i=1}^d p_i \frac{\partial f}{\partial x_i}(x_1, \dots, x_d). \tag{4.3}$$

(1) The numerator in the definition of G_p is homogeneous of degree $r+1$ in $x_0, \dots, x_d$, so the quotient by x_0 is homogeneous of degree r. Since x_{d+1} has weight r, G_p is a weighted homogeneous polynomial of degree r, and $G_p \in H^0(Z, \mathcal{O}_Z(r))$. Since $Z \cap (x_0 \neq 0)$ is the graph of the function $f(x_1/x_0, \dots, x_d/x_0)$ in the chart $(x_0 \neq 0) \cong \mathbb{A}^{d+1}$ with coordinates x_j/x_0,

$1 \le j \le d$, and x_{d+1}/x_0^r, we have $Z \cap (x_0 \neq 0) \cong \mathbb{A}^d$ with coordinates $x_1/x_0, \dots, x_d/x_0$. Since f is homogeneous of degree $r+1$, (4.2) implies that

$$\frac{G_p}{x_0^r} = f\Big(\frac{x_1}{x_0} - p_1, \dots, \frac{x_d}{x_0} - p_d\Big)$$

on $Z \cap (x_0 \neq 0)$, and the right-hand side is a non-zero polynomial function on $\mathbb{A}^d$, since $f \neq 0$. Since Z is integral and $\mathcal{O}_Z(r)$ is an ample invertible sheaf (Lemma 4.2(1)), $D_p \ge 0$ is a Cartier divisor with $\mathcal{O}_Z(D_p) \cong \mathcal{O}_Z(r)$, hence ample. Since f is homogeneous of degree $r+1 \ge 2$, $f(p-p) = f(0) = 0$, so $p_Z \in D_p$. Near o, $\mathcal{O}_Z(r)$ is generated by x_{d+1}, and

$$\frac{G_p}{x_{d+1}} = 1 + \frac{f(x_1 - p_1x_0, \dots, x_d - p_dx_0) - f(x_1, \dots, x_d)}{x_0x_{d+1}},$$

and the second term is $(G_p - x_{d+1})/x_{d+1}$. Since $G_p - x_{d+1}$ is a homogeneous polynomial of degree $r \ge 1$ in $x_0, \dots, x_d$ by Construction 4.6, $(G_p - x_{d+1})/x_{d+1}$ is a regular function on $Z \cap (x_{d+1} \neq 0)$ that vanishes at o. Thus $o \notin D_p$.

(2) Let $u = (u_1, \dots, u_d) \in \mathbb{C}^d \backslash \{0\}$ such that $f(u) = 0$. The polynomial $f(py_0 + uy_1) = f(p_1y_0 + u_1y_1, \dots, p_dy_0 + u_dy_1)$ is homogeneous of degree $r+1$ in y_0, y_1 and equals $y_1^{r+1}f(u) = 0$ after substituting $y_0 = 0$. Hence $f(py_0 + uy_1)/y_0$ is a homogeneous polynomial of degree r, and we let $\gamma_u : \mathbb{P}^1 \to \mathbb{P}(1^{d+1}, r)$ be

$$[y_0 : y_1] \to [y_0 : p_1y_0 + u_1y_1 : \cdots : p_dy_0 + u_dy_1 : f(py_0 + uy_1)/y_0].$$

Since $u \neq 0$, the first $d+1$ coordinates of γ_u do not vanish simultaneously: $y_0 = 0$ and $y_1u = 0$ imply $y_0 = y_1 = 0$. Thus γ_u is a morphism whose image does not contain o. Since x_0x_{d+1} pulls back to $f(py_0 + uy_1)$, which is the pullback of $f(x_1, \dots, x_d)$, the image of γ_u is contained in Z. Let $C := \gamma_u(\mathbb{P}^1)$. The composite

$$\pi \circ \gamma_u : [y_0 : y_1] \to [y_0 : p_1y_0 + u_1y_1 : \cdots : p_dy_0 + u_dy_1]$$

is induced by the injective linear map $(y_0, y_1) \to (y_0, py_0 + uy_1)$, since $(1, p)$ and $(0, u)$ are linearly independent. Hence $\pi \circ \gamma_u$ is an isomorphism onto a line of $\mathbb{P}^d$, and $\gamma_u : \mathbb{P}^1 \to C$ is an isomorphism with inverse $(\pi \circ \gamma_u)^{-1} \circ \pi|_C$. In particular, C is a smooth rational curve and $C \subset Z \backslash \{o\}$. Since $\mathcal{O}_Z(1)$ is generated by x_i on $Z \cap (x_i \neq 0)$ for $0 \le i \le d$, we have

$$\mathcal{O}_Z(1)|_{Z\backslash\{o\}} \cong (\pi|_{Z\backslash\{o\}})^*\mathcal{O}_{\mathbb{P}^d}(1)$$

by Lemma 4.2(1). Hence

$$(H \cdot C) = \deg \gamma_u^*\mathcal{O}_Z(1) = \deg(\pi \circ \gamma_u)^*\mathcal{O}_{\mathbb{P}^d}(1) = 1.$$

Moreover, $\gamma_u([1:0]) = [1 : p_1 : \cdots : p_d : f(p)] = p_Z$, so $p_Z \in C$. The pullback of $x_i - p_ix_0$ by γ_u is u_iy_1 for $1 \le i \le d$. Since the right-hand side of (4.2) vanishes on $C \subset Z$, (4.2) implies that

$$y_0 \cdot \gamma_u^*G_p = f(u_1y_1, \dots, u_dy_1) = y_1^{r+1}f(u) = 0$$

in $\mathbb{C}[y_0, y_1]$. Thus $\gamma_u^*G_p = 0$, and $C \subset D_p$.

It remains to show that any closed point $z \in D_p \backslash \{p_Z\}$ is contained in $\gamma_u(\mathbb{P}^1)$ for some $u \in \mathbb{C}^d \backslash \{0\}$ such that $f(u) = 0$. We consider two cases.

Case 1. $x_0(z) \neq 0$. Then $z = [1 : w_1 : \cdots : w_d : f(w)]$ for some $w = (w_1, \dots, w_d) \in \mathbb{C}^d$, and $w \neq p$ since $z \neq p_Z$. By (4.2), $z \in D_p$ implies that $f(w - p) = 0$. Let $u := w - p$. Then $u \neq 0$, $f(u) = 0$, and $\gamma_u([1:1]) = [1 : w_1 : \cdots : w_d : f(w)] = z$.

Case 2. $x_0(z) = 0$. Since $o \notin D_p$ by (1), $z = [0 : u_1 : \cdots : u_d : \beta]$ for some $u = (u_1, \dots, u_d) \in \mathbb{C}^d \backslash \{0\}$ and $\beta \in \mathbb{C}$. Since $z \in Z$, $f(u) = 0$. Since $z \in D_p$, (4.3) implies that

$$\beta = \sum_{i=1}^{d} p_i \frac{\partial f}{\partial x_i}(u).$$

The coefficient of y_0 in $f(py_0+u)$, viewed as a polynomial in y_0, is $\sum_{i=1}^d p_i \frac{\partial f}{\partial x_i}(u)$, and $f(u)=0$. Hence the last coordinate of $\gamma_u([0:1])$ is β, and $\gamma_u([0:1])=z$.

(3) By Lemma 4.2(2), any singular point of Z other than o is $z=[0:u_1:\cdots:u_d:0]$ for some $u\neq 0$ such that $\frac{\partial f}{\partial x_i}(u)=0$ for $1\leq i\leq d$. By (4.3), $G_p(z)=0$, so $z\in D_p$. □

By [Zhu18, Lemma 3.5], if x is a smooth point of a normal projective variety X, $W\subset|L|$ is a linear system without base point at x for a $\mathbb{Q}$-Cartier divisor L, and C is an irreducible curve through x that intersects the base locus of W, then $s(W,x)<(L\cdot C)/\operatorname{mult}_x C$. When $\operatorname{mult}_x C=1$ and $(L\cdot C)$ is an integer, the jet order thus drops by at least 1, and a drop by 1 in each degree k does not lower ϵ_{mov}. In Proposition 4.8, Izumi's theorem [Izu85, Ree89] turns vanishing of order at least kt along a prime divisor E over Z into vanishing of order at least kt/c at a closed point of $T\cap D_p$, so that the drop on the curves of Lemma 4.7(2) grows linearly in k. Since D_p does not contain o, the point o is not excluded in this way, and we treat o with normalized volumes in Section 6.

Proposition 4.8. *Notation and conditions as in Set-up 4.1 and Construction 4.6. Let E be a prime divisor over Z, $T:=\operatorname{center}_Z(\operatorname{ord}_E)$, $t>0$ a real number, and $V_\bullet=(V_k)_{k\geq 1}$ a sequence of linear subspaces $V_k\subset H^0(Z,\mathcal{O}_Z(dk))$ (we regard $V_\bullet$ as a sequence for the Weil divisor dH in Definition 2.4 and Lemma 2.8, since $\mathcal{O}_Z(dk)\cong\mathcal{O}_Z(dkH)$: both sheaves are reflexive of rank one, see the proof of Lemma 5.5, and they coincide on $Z\backslash\{o\}$ by Lemma 4.2(1)) such that $V_k\subset H^0(Z,\mathfrak{a}_{kt}(\operatorname{ord}_E)\cdot\mathcal{O}_Z(dk))$ for any $k\geq 1$, where $\mathfrak{a}_{kt}(\operatorname{ord}_E)\cdot\mathcal{O}_Z(dk)$ is the image of $\mathfrak{a}_{kt}(\operatorname{ord}_E)\otimes\mathcal{O}_Z(dk)\to\mathcal{O}_Z(dk)$. Then the following hold.*

(1) Let $p\in\mathbb{C}^d$ such that $p_Z\notin T$, and let $z\in T\cap D_p$ be a closed point. Then there exists a real number $c>0$ such that $s(V_k,p_Z)\leq dk-\lceil kt/c\rceil$ for any $k\geq 1$ with $s(V_k,p_Z)\geq 0$.

(2) If $\epsilon_{\mathrm{mov}}(V_\bullet,x)\geq d$ for a very general point $x\in Z$, then T is a closed point of $Z_{\mathrm{reg}}\cup\{o\}$.

Proof. The proof is long, so we divide it into several steps. In **Steps 1 and 2**, we fix $p\in\mathbb{C}^d$ and a closed point $z\in T\cap D_p$ such that $p_Z\notin T$.

Step 1. In this step we bound ord_E on $\mathcal{O}_{Z,z}$ by a multiple of the $\mathfrak{m}_z$-adic order.

For $h\in\mathcal{O}_{Z,z}\backslash\{0\}$, let $\operatorname{ord}_{\mathfrak{m}_z}(h):=\max\{j\mid h\in\mathfrak{m}_z^j\}$, which is finite by the Krull intersection theorem. We show that there exists a real number $c>0$ such that

$$\operatorname{ord}_E(h)\leq c\operatorname{ord}_{\mathfrak{m}_z}(h)\quad\text{for any } h\in\mathcal{O}_{Z,z}\backslash\{0\}. \tag{4.4}$$

By [KM98, Lemma 2.45] and [KM98, Theorem 0.2], there exists a projective birational morphism $g:Z'\to Z$ from a smooth variety Z' such that E is a smooth prime divisor on Z'. Since $g(E)=T$ and $z\in T$, there exists a closed point $w\in E$ such that $g(w)=z$. Let e be a local equation of E at w. Since E is smooth at w, e is a prime element of $\mathcal{O}_{Z',w}$. Let $h\in\mathcal{O}_{Z,z}\backslash\{0\}$. Since $\mathcal{O}_{Z',w}$ is a UFD [Mat86, Theorem 20.3], we may write $g^*h=e^mh'$ for some integer $m\geq 0$ and some $h'\in\mathcal{O}_{Z',w}\backslash e\mathcal{O}_{Z',w}$. Since $\mathcal{O}_{Z',E}$ is the localization of $\mathcal{O}_{Z',w}$ at the prime ideal $e\mathcal{O}_{Z',w}$, h' is a unit in $\mathcal{O}_{Z',E}$, so $m=\operatorname{ord}_E(h)$. Since w is a smooth point of Z', $\operatorname{ord}_w(h'')=\max\{j\mid h''\in\mathfrak{m}_w^j\}$ for any $h''\in\mathcal{O}_{Z',w}\backslash\{0\}$, and $e\in\mathfrak{m}_w$. Hence

$$\operatorname{ord}_w(g^*h)\geq\operatorname{ord}_E(h).$$

The valuation ord_w of $\mathbb{C}(Z')=\mathbb{C}(Z)$ is divisorial and $\operatorname{center}_Z(\operatorname{ord}_w)=\{z\}$. Since Z is normal and ord_w is a divisorial valuation centered at z, Izumi's theorem [Izu85, Ree89], in the form of [BFJ14, Introduction, Izumi's Theorem] (for a closed point z of a normal variety and a divisorial valuation v centered at z, there exists $c>0$ such that $v(h)\leq c\operatorname{ord}_{\mathfrak{m}_z}(h)$ for any $h\in\mathcal{O}_{Z,z}\backslash\{0\}$), implies that there exists a real number $c>0$ such that $\operatorname{ord}_w(g^*h)\leq c\operatorname{ord}_{\mathfrak{m}_z}(h)$ for any $h\in\mathcal{O}_{Z,z}\backslash\{0\}$, also when z is a singular point of Z. Thus (4.4) holds.

In particular, any $h\in\mathcal{O}_{Z,z}$ such that $\operatorname{ord}_E(h)\geq kt$ satisfies $\operatorname{ord}_{\mathfrak{m}_z}(h)\geq kt/c$, hence

$$\mathfrak{a}_{kt}(\operatorname{ord}_E)\cdot\mathcal{O}_{Z,z}\subset\mathfrak{m}_z^{\lceil kt/c\rceil}\quad\text{for any } k\geq 1. \tag{4.5}$$

Step 2. In this step we prove (1), using a curve through p_Z and z.

On $Z\cap(x_i\neq 0)$, $0\leq i\leq d$, the sheaf $\mathcal{O}_Z(m)$ is generated by x_i^m for any integer m, hence $\mathcal{O}_Z(m)\cong\mathcal{O}_Z(1)^{\otimes m}$ on $Z\backslash\{o\}$ by Lemma 4.2(1). Since $p_Z\notin T$ and $z\in T$, $z\in D_p\backslash\{p_Z\}$. By Lemma 4.7(2), there exists a smooth rational curve $C\subset D_p$ such that $p_Z,z\in C$ and $(H\cdot C)=1$. By Lemma 4.7(1), $o\notin D_p$, so $C\subset Z\backslash\{o\}$. Thus $\mathcal{O}_Z(dk)$ is invertible near C, $\mathfrak{a}_{kt}(\mathrm{ord}_E)\cdot\mathcal{O}_Z(dk)$ is the product of an ideal sheaf and an invertible sheaf near C, and

$$\mathcal{O}_Z(dk)|_C\cong\mathcal{O}_{\mathbb{P}^1}(dk)$$

for any $k\geq 1$, since $(H\cdot C)=1$.

Let $k\geq 1$ such that $s:=s(V_k,p_Z)\geq 0$. Here $p_Z\in Z_{\mathrm{reg}}$ by Lemma 4.2(2), so $s(V_k,p_Z)$ is defined. Let λ be a local generator of $\mathcal{O}_Z(dk)$ at z. Any $\sigma\in V_k$ is $h\lambda$ near z for some $h\in\mathfrak{a}_{kt}(\mathrm{ord}_E)\cdot\mathcal{O}_{Z,z}$. Recall that $c>0$ is the real number in (4.4). By (4.5), $h\in\mathfrak{m}_z^{\lceil kt/c\rceil}$. Since $\mathfrak{m}_z\mathcal{O}_{C,z}\subset\mathfrak{m}_{C,z}$, the restriction $\sigma|_C$ is either zero or vanishes to order at least $\lceil kt/c\rceil$ at z.

Since C is a closed subvariety of Z and both are smooth at p_Z, there exists $\zeta\in\mathfrak{m}_{p_Z}$ such that $\zeta|_C$ is a uniformizer of $\mathcal{O}_{C,p_Z}$. Let λ' be a local generator of $\mathcal{O}_Z(dk)$ at p_Z. Since the evaluation map $V_k\to\mathcal{O}_Z(dk)\otimes\mathcal{O}_Z/\mathfrak{m}_{p_Z}^{s+1}$ is surjective, there exists $\sigma\in V_k$ such that $\sigma-\zeta^s\lambda'\in\mathfrak{m}_{p_Z}^{s+1}\lambda'$ near p_Z. Then $\sigma|_C\neq 0$, and $\sigma|_C$ vanishes to order exactly s at p_Z. Since $z\neq p_Z$ and $\sigma|_C$ is a non-zero section of $\mathcal{O}_Z(dk)|_C\cong\mathcal{O}_{\mathbb{P}^1}(dk)$, we obtain (cf. [Zhu18, Lemma 3.5])

$$s(V_k,p_Z)+\lceil kt/c\rceil\leq\deg\mathcal{O}_Z(dk)|_C=(dkH\cdot C)=dk.$$

This proves (1).

Step 3. In this step we prove (2).

Assume that $\epsilon_{\mathrm{mov}}(V_\bullet,x)\geq d$ for a very general point $x\in Z$. In particular, $\epsilon_{\mathrm{mov}}(V_\bullet,x)\geq d$ for some $x\in Z_{\mathrm{reg}}$. By Lemma 2.8, there exists a subset $U\subset Z_{\mathrm{reg}}$ that is the intersection of countably many dense open subsets of Z such that

$$\epsilon_{\mathrm{mov}}(V_\bullet,y)=\sup_{y'\in Z_{\mathrm{reg}}}\epsilon_{\mathrm{mov}}(V_\bullet,y')\geq d$$

for any $y\in U$. Since ord_E is a non-trivial valuation of $\mathbb{C}(Z)$, ord_E is not centered at the generic point of Z, so $T\neq Z$. Since $\mathbb{C}$ is uncountable, the intersection of U with the dense open subset $(Z\cap(x_0\neq 0))\backslash T$ is non-empty. Since any point of $Z\cap(x_0\neq 0)$ is of the form p_Z, there exists $p\in\mathbb{C}^d$ such that $p_Z\in U$ and $p_Z\notin T$.

We show that $T\cap D_p=\emptyset$. Suppose that $T\cap D_p\neq\emptyset$, and let $z\in T\cap D_p$ be a closed point. Let $c>0$ be the real number of (1) for p and z. For any $k\geq 1$, either $s(V_k,p_Z)=-1$, or $s(V_k,p_Z)\leq dk-\lceil kt/c\rceil\leq dk-kt/c$ by (1). Thus

$$\epsilon_{\mathrm{mov}}(V_\bullet,p_Z)=\limsup_{k\to+\infty}\frac{s(V_k,p_Z)}{k}\leq\max\left\{0,d-\frac{t}{c}\right\}<d,$$

which contradicts $p_Z\in U$. Therefore $T\cap D_p=\emptyset$.

Suppose that $\dim T\geq 1$. Since $T\cap D_p=\emptyset$, $\mathcal{O}_Z(D_p)|_T\cong\mathcal{O}_T$, so $(D_p^{\dim T}\cdot T)=0$. This contradicts the ampleness of the Cartier divisor D_p (Lemma 4.7(1)) and [KM98, Theorem 1.37]. Thus $\dim T=0$, and $T=\{z_0\}$ for a closed point $z_0\in Z$. If $z_0\neq o$ and z_0 is a singular point of Z, then $z_0\in D_p$ by Lemma 4.7(3), which contradicts $T\cap D_p=\emptyset$. Hence $z_0\in Z_{\mathrm{reg}}\cup\{o\}$. This proves (2). □

5. Asymptotic base ideals on weighted hypersurfaces

In this section, we regard the anti-canonical sections of a variety Y birational to Z as a graded linear series on Z, and we prove a bound at a point for the asymptotic base ideals of the series by Nadel vanishing (Proposition 5.3). Proposition 5.3 is a local form of the comparison of volumes in [Zhu18, Lemma 2.15], which here only yields $\mathrm{vol}(-K_Y)<\mathrm{vol}(-K_Z)$.

Set-up 5.1. Notation and conditions as in Set-up 4.1. Let Y, g, and I be as follows.

(1) Z is klt.

(2) Y is a $\mathbb{Q}$-factorial terminal projective variety of dimension d such that $\epsilon_{\mathrm{mov}}(-K_Y) = d$ and $\mathrm{vol}(-K_Y) = d^d$.
(3) $g : Y \dashrightarrow Z$ is a birational map such that g^{-1} does not contract any divisor.
(4) I is a positive integer such that IK_Y and IK_Z are Cartier and $H^0(Y, \mathcal{O}_Y(-IK_Y)) \neq 0$.

Such an integer I exists, since $\mathrm{vol}(-K_Y) > 0$ and hence $-K_Y$ is big.

Construction 5.2. Notation and conditions as in Set-up 5.1. For any integer $m \geq 1$, we let $V_m := H^0(Y, \mathcal{O}_Y(-mIK_Y))$, regarded as a linear subspace of $H^0(Z, \mathcal{O}_Z(-mIK_Z))$ by Lemma 2.10(1). Then $V_\bullet = (V_m)_{m\geq 1}$ is a graded linear series of $-IK_Z$, since $V_m V_l \subset V_{m+l}$ in $\mathbb{C}(Y) = \mathbb{C}(Z)$ for any $m, l \geq 1$. We let $\mathfrak{b}_\bullet = (\mathfrak{b}_m)_{m\geq 1}$, where $\mathfrak{b}_m$ is the base ideal of V_m on Z. We let $\mathfrak{c}_\bullet = (\mathfrak{c}_m)_{m\geq 1}$, where $\mathfrak{c}_m$ is the base ideal of $H^0(Y, \mathcal{O}_Y(-mIK_Y))$ on Y. Since $V_1 \neq 0$, $\mathfrak{b}_\bullet$ and $\mathfrak{c}_\bullet$ are graded sequences of ideals. For any $v \in \mathrm{Val}_Z$, we let $\sigma(v) := v(\mathfrak{b}_\bullet)/I$. For any $v \in \mathrm{Val}_Z$, which has a center on Y since Y is projective, we let $\sigma_Y(v) := v(\mathfrak{c}_\bullet)/I$. We let $\tau := I\,\mathrm{lct}(Z; \mathfrak{b}_\bullet)$. For any integer $k \geq 1$, we let $\mathcal{J}_k := \mathcal{J}(Z, k \cdot \mathfrak{b}_\bullet)$.

We use $V_\bullet$ rather than the complete linear series of $-IK_Z$, since for any $m \geq 1$, the members of $|V_m|$ are exactly the divisors on Z whose corresponding divisors on Y are members of $|-mIK_Y|$. Thus a klt boundary on Z built from members of $|V_m|$ can be carried back to Y (Lemma 6.2), and τ is the threshold for such boundaries to be klt. We also use $\mathfrak{c}_\bullet$ and σ_Y, which measure the anti-canonical sections of Y along valuations on Y, are compared with $\mathfrak{b}_\bullet$ and σ in Lemma 6.4, and are used in Lemma 6.6 to extract the prime divisor Γ of Set-up 4.1 over Y without losing sections.

The constant on the right-hand side of the inequality in Proposition 5.3 below is the difference of $\mathrm{vol}(-K_Z)$ and $\mathrm{vol}(-K_Y)$. We use the asymptotic multiplier ideals $\mathcal{J}_k$ rather than the base ideals $\mathfrak{b}_k$, since Nadel vanishing holds for $\mathcal{J}_k$.

Proposition 5.3. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Let $v \in \mathrm{Val}_Z$ such that $a(v, Z) < +\infty$ and $\sigma(v) > 0$. Then $\mathrm{center}_Z v$ is a closed point, and*

$$\sigma(v)^d \,\mathrm{vol}(v) \leq \frac{d^d}{r}.$$

We prove Proposition 5.3 at the end of Subsection 5.2.

5.1. **The image series of anti-canonical sections.** In this subsection, we show that the base ideals of $V_\bullet$ are non-zero, and we compute the asymptotic jet orders and the volume of $V_\bullet$.

Lemma 5.4. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Then $\mathfrak{b}_m \neq 0$ for any $m \geq 1$, $\epsilon_{\mathrm{mov}}(V_\bullet, x) = Id$ for a very general point $x \in Z$, and*

$$\lim_{m\to+\infty} \frac{d!\,\dim V_m}{m^d} = I^d d^d.$$

Proof. Since $V_1 \neq 0$ by Set-up 5.1(4), $\mathfrak{b}_1 \neq 0$. For any $m \geq 1$, the products of m elements of V_1 belong to V_m, so $\mathfrak{b}_1^m \subset \mathfrak{b}_m$. Thus $\mathfrak{b}_m \neq 0$.

Let $U \subset Y_{\mathrm{reg}}$ be a non-empty open subset such that $g|_U$ is an open immersion and $g(U) \subset Z_{\mathrm{reg}}$. For any $y \in U$ and any $m \geq 1$, Lemma 2.10(2) with $k = mI$ implies

$$s(V_m, g(y)) = s(H^0(Y, \mathcal{O}_Y(-mIK_Y)), y),$$

where on the left-hand side V_m is regarded as a subspace of $H^0(Z, \mathcal{O}_Z(-mIK_Z))$. By Lemma 2.9 for the graded linear series $(H^0(Y, \mathcal{O}_Y(-mK_Y)))_{m\geq 1}$ of $-K_Y$,

$$\epsilon_{\mathrm{mov}}(V_\bullet, g(y)) = I\epsilon_{\mathrm{mov}}(-K_Y, y)$$

for any $y \in U$.

Let $y \in U$ be a very general point of Y. Since $g|_U$ is an open immersion, $g(y)$ is a very general point of Z. By Lemma 2.8 applied to $-K_Y$ and to $V_\bullet$,

$$\epsilon_{\mathrm{mov}}(V_\bullet, x) = \epsilon_{\mathrm{mov}}(V_\bullet, g(y)) = I\epsilon_{\mathrm{mov}}(-K_Y, y) = I\epsilon_{\mathrm{mov}}(-K_Y) = Id$$

for a very general point $x \in Z$.

Finally, $\dim V_m = h^0(Y, \mathcal{O}_Y(-mIK_Y))$ for any $m \geq 1$. By Theorem 2.6 applied to $-IK_Y$,

$$\lim_{m\to+\infty} \frac{d! \dim V_m}{m^d} = \mathrm{vol}(-IK_Y) = I^d \,\mathrm{vol}(-K_Y) = I^d d^d.$$

□

5.2. **A local bound for asymptotic orders.** In this subsection, we prove Proposition 5.3.

Proposition 4.8 concerns divisorial valuations, whereas the valuations computing log canonical thresholds in Section 6 are only real valuations. In the next lemma, we approximate such a valuation v by a divisorial valuation w with $\mathrm{center}_Z w = \mathrm{center}_Z v$.

Lemma 5.5. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Let $v \in \mathrm{Val}_Z$ such that $a(v, Z) < +\infty$ and $\sigma(v) > 0$. Then $\mathrm{center}_Z v$ is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$.*

Proof. Since $v(\mathfrak{b}_\bullet) = I\sigma(v) > 0$, Theorem 2.15(3) implies that there exists an open subset $\mathcal{U} \subset \mathrm{Val}_Z$ containing v such that $w(\mathfrak{b}_\bullet) > 0$ for any $w \in \mathcal{U}$ with $a(w, Z) < +\infty$.

Let $h : Z' \to Z$ be a resolution of singularities such that Z' is projective. Then $\mathrm{Val}_{Z'} = \mathrm{Val}_Z$, since Z and Z' are projective. For any non-zero ideal sheaf $\mathfrak{a}$ on Z and any $w \in \mathrm{Val}_Z$, we have $w(\mathfrak{a}) = w(\mathfrak{a} \cdot \mathcal{O}_{Z'})$, since local generators of $\mathfrak{a}$ at the generic point of $\mathrm{center}_Z w$ generate $\mathfrak{a} \cdot \mathcal{O}_{Z'}$ at the generic point of $\mathrm{center}_{Z'} w$, and $w \geq 0$ on both local rings (Definition 2.12(2)). Hence $\mathcal{U}$ is open in the weakest topology on $\mathrm{Val}_{Z'}$ such that $w \to w(\mathfrak{a}')$ is continuous for any non-zero ideal sheaf $\mathfrak{a}'$ on Z', which is the topology on $\mathrm{Val}_{Z'}$ in [JM12] by [JM12, Lemma 4.1]. In particular, $\mathcal{U}$ is an open subset of $\mathrm{Val}_{Z'}$. We show that there exists a divisorial valuation $w \in \mathcal{U}$ such that $\mathrm{center}_{Z'} w = \mathrm{center}_{Z'} v$.

By [JM12, Corollary 4.10], there exists a log smooth pair (Z'', D) over Z' such that the image of D in Z' is contained in $\mathrm{center}_{Z'} v$, and the image v' of v under the retraction of [JM12, Corollary 4.8] onto the valuations monomial in (Z'', D) belongs to $\mathcal{U}$. Since v' is quasi-monomial, $a(v', Z') < +\infty$ [JM12, Proposition 5.1], hence $a(v', Z) < +\infty$ [BJ20, Section 1.5], so $v'(\mathfrak{b}_\bullet) > 0$. In particular, v' is non-trivial, and $\mathrm{center}_{Z''} v'$ is contained in D. Hence $\mathrm{center}_{Z'} v' \subset \mathrm{center}_{Z'} v$. By [JM12, Corollary 4.8], $v'(\mathfrak{a}) \leq v(\mathfrak{a})$ for any ideal sheaf $\mathfrak{a}$ on Z'. If $\mathrm{center}_{Z'} v \not\subset \mathrm{center}_{Z'} v'$, then the ideal sheaf $\mathfrak{a}$ of $\mathrm{center}_{Z'} v'$ satisfies $v'(\mathfrak{a}) > 0 = v(\mathfrak{a})$, which is not possible. Thus $\mathrm{center}_{Z'} v' = \mathrm{center}_{Z'} v$.

Let η' be the generic point of $\mathrm{center}_{Z''} v'$, and let $D_1, \ldots, D_p$ be the components of D containing η'. The valuations monomial in (Z'', D) at η' with positive weights on $D_1, \ldots, D_p$ have center $\mathrm{center}_{Z''} v'$ on Z'', and such a valuation is divisorial if its weights are rational [JM12, Remark 3.9]. The map from the set of valuations monomial in (Z'', D) at η' to $\mathbb{R}^p_{\geq 0}$, $w' \to (w'(D_i))_i$, is a homeomorphism [JM12, Lemma 4.5], and v' has positive weights since its center is η' (cf. [JM12, Proposition 3.1]). Since $\mathbb{Q}^p_{>0}$ is dense in $\mathbb{R}^p_{>0}$, choosing rational weights close to the weights of v', we obtain a divisorial valuation $w \in \mathcal{U}$ such that $\mathrm{center}_{Z''} w = \mathrm{center}_{Z''} v'$. In particular, $\mathrm{center}_{Z'} w = \mathrm{center}_{Z'} v$.

Since w is divisorial, $a(w, Z) < +\infty$, so $\sigma(w) > 0$. Moreover,

$$\mathrm{center}_Z w = h(\mathrm{center}_{Z'} w) = h(\mathrm{center}_{Z'} v) = \mathrm{center}_Z v.$$

Since $\sigma(cw) = c\sigma(w)$ and $\mathrm{center}_Z(cw) = \mathrm{center}_Z w$ for any real number $c > 0$, possibly replacing w with a positive multiple, we may assume that $w = \mathrm{ord}_E$ for a prime divisor E over Z. Let $t := \sigma(w) > 0$.

For any integer $\ell \geq 1$, $-\ell K_Z \sim d\ell H$ by Lemma 4.2(4). The sheaves $\mathcal{O}_Z(d\ell H)$ and $\mathcal{O}_Z(d\ell)$ are reflexive of rank one. For $\mathcal{O}_Z(d\ell)$, this holds since it is invertible on $Z \backslash \{o\}$ by Lemma 4.2(1), and on $Z \cap (x_{d+1} \neq 0) \cong \mathbb{A}^d/\mu_r$ (Lemma 4.2(3)) it is the sheaf of μ_r-semi-invariant functions of weight $d\ell$ modulo r, a direct summand of the push-forward of $\mathcal{O}_{\mathbb{A}^d}$. The sheaves $\mathcal{O}_Z(d\ell H)$ and $\mathcal{O}_Z(d\ell)$ are isomorphic on $Z \backslash \{o\}$ by Lemma 4.2(1) and $\mathcal{O}_Z(H) \cong \mathcal{O}_Z(1)$. Since $\mathrm{codim}_Z\{o\} = d \geq 2$, there exists an isomorphism $\phi_\ell : \mathcal{O}_Z(-\ell K_Z) \to \mathcal{O}_Z(d\ell)$. For any ideal sheaf $\mathfrak{a}$ on Z, ϕ_ℓ maps $\mathfrak{a} \cdot \mathcal{O}_Z(-\ell K_Z)$ onto $\mathfrak{a} \cdot \mathcal{O}_Z(d\ell)$. Moreover, $s(\phi_\ell(V), x) = s(V, x)$ for any $x \in Z_{\mathrm{reg}}$ and any linear

subspace $V \subset H^0(Z, \mathcal{O}_Z(-\ell K_Z))$. We let $W_\ell := \phi_\ell(V_m)$ if $\ell = mI$ for an integer $m \geq 1$, and $W_\ell := 0$ otherwise. Then $W_\ell \subset H^0(Z, \mathcal{O}_Z(d\ell))$ for any $\ell \geq 1$.

We check that $W_\bullet = (W_\ell)_{\ell \geq 1}$ satisfies the conditions of Proposition 4.8(2) with respect to E and t. Let $\ell = mI$. By the definition of $v(\mathfrak{b}_\bullet)$ in Definition 2.14,

$$\operatorname{mult}_E \mathfrak{b}_m = w(\mathfrak{b}_m) \geq mw(\mathfrak{b}_\bullet) = mI\sigma(w) = \ell t,$$

so $\mathfrak{b}_m \subset \mathfrak{a}_{\ell t}(\operatorname{ord}_E)$. Since $V_m \subset H^0(Z, \mathfrak{b}_m \cdot \mathcal{O}_Z(-\ell K_Z))$, we have

$$W_\ell \subset H^0(Z, \mathfrak{a}_{\ell t}(\operatorname{ord}_E) \cdot \mathcal{O}_Z(d\ell)).$$

If $I \nmid \ell$, then $W_\ell = 0 \subset H^0(Z, \mathfrak{a}_{\ell t}(\operatorname{ord}_E) \cdot \mathcal{O}_Z(d\ell))$. For any $x \in Z_{\mathrm{reg}}$, $s(W_{mI}, x) = s(V_m, x)$ for any $m \geq 1$, so

$$\epsilon_{\mathrm{mov}}(W_\bullet, x) \geq \limsup_{m \to +\infty} \frac{s(V_m, x)}{mI} = \frac{1}{I} \epsilon_{\mathrm{mov}}(V_\bullet, x).$$

By Lemma 5.4, $\epsilon_{\mathrm{mov}}(W_\bullet, x) \geq d$ for a very general point $x \in Z$. Thus Proposition 4.8(2) implies that $\operatorname{center}_Z(\operatorname{ord}_E)$ is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$. Since $\operatorname{center}_Z(\operatorname{ord}_E) = \operatorname{center}_Z v$, Lemma 5.5 follows. □

Nadel vanishing only bounds the dimension of the space of global sections of $\mathcal{O}_Z/\mathcal{J}_k$ twisted by a line bundle. To read off the length of $\mathcal{O}_{Z,q}/(\mathcal{J}_k)_q$ at a point q, we need q to be an isolated point of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$.

Lemma 5.6. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Let $k \geq 1$ be an integer and $q \in Z_{\mathrm{reg}} \cup \{o\}$ a closed point such that $q \in \operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$. Then q is an isolated point of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$.*

Proof. By Lemma 2.17(1) and Lemma 5.4, $\mathcal{J}_k \supset \mathfrak{b}_k \neq 0$. We first show that for any divisorial valuation $w \in \operatorname{Val}_Z$ such that the generic point ξ of $\operatorname{center}_Z w$ belongs to Z_{reg}, we have $w(\mathcal{J}_k) \leq kI\sigma(w)$. By Lemma 2.17(3), $\mathfrak{b}_m^k \cdot \mathcal{O}_{Z,\xi} \subset \mathcal{J}_k^m \cdot \mathcal{O}_{Z,\xi}$ for any $m \geq 1$, so $kw(\mathfrak{b}_m) \geq mw(\mathcal{J}_k)$. Taking the infimum over m, we obtain

$$w(\mathcal{J}_k) \leq kw(\mathfrak{b}_\bullet) = kI\sigma(w). \tag{5.1}$$

Suppose that T is a positive-dimensional irreducible component of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$ containing q. Let η be the generic point of T. If $q \in Z_{\mathrm{reg}}$, then $\eta \in Z_{\mathrm{reg}}$ since Z_{reg} is an open neighborhood of q. If $q = o$, then $\eta \in Z_{\mathrm{reg}}$ since $Z \cap (x_{d+1} \neq 0)$ is an open neighborhood of o and $(Z \cap (x_{d+1} \neq 0))\backslash\{o\} \subset Z_{\mathrm{reg}}$ (Lemma 4.2(3)). There exists a divisorial valuation $w \in \operatorname{Val}_Z$ such that $\operatorname{center}_Z w = T$: for instance, let $w = \operatorname{ord}_F$, where F is an irreducible component dominating T of the inverse image of T in the normalization of the blow-up of Z along T. Since $\eta \in \operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$, $\mathcal{J}_k \cdot \mathcal{O}_{Z,\eta}$ is contained in the maximal ideal of $\mathcal{O}_{Z,\eta}$, so $w(\mathcal{J}_k) > 0$. Since $\eta \in Z_{\mathrm{reg}}$, applying (5.1) to $w = \operatorname{ord}_F$, we obtain $kI\sigma(w) \geq w(\mathcal{J}_k) > 0$, hence $\sigma(w) > 0$. Since $a(w, Z) < +\infty$, Lemma 5.5 implies that $T = \operatorname{center}_Z w$ is a closed point, a contradiction.

Therefore any irreducible component of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$ containing q is equal to $\{q\}$, so q is an isolated point of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$. □

Proof of Proposition 5.3. By Lemma 5.5, $q := \operatorname{center}_Z v$ is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$. Let $L := -IK_Z$. By Lemma 4.2(4), Z is a $\mathbb{Q}$-Fano variety, so L is an ample Cartier divisor, and

$$(L^d) = I^d((-K_Z)^d) = I^d d^d(H^d) = \frac{I^d d^d(r+1)}{r}.$$

Moreover, $V_\bullet$ is a graded linear series of L with $V_1 \neq 0$, so Z, L, and $V_\bullet$ satisfy the hypotheses of Lemma 2.17.

For any integer $k \geq 1$, let $t_k := kI\sigma(v) - a(v, Z)$. Let $k \geq 1$ be an integer such that $t_k > 0$. Since $v(\mathfrak{b}_\bullet) = I\sigma(v)$, Lemma 2.17(4) implies

$$\mathcal{J}_k \subset \mathfrak{a}_{t_k}(v).$$

Since $t_k > 0$ and $\operatorname{center}_Z v = q$, $\mathfrak{a}_{t_k}(v) \cdot \mathcal{O}_{Z,q}$ is contained in the maximal ideal of $\mathcal{O}_{Z,q}$. Thus $q \in \operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$, and q is an isolated point of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$ by Lemma 5.6.

By Lemma 2.17(1), $\mathfrak{b}_k \subset \mathcal{J}_k$, so

$$V_k \subset H^0(Z, \mathcal{O}_Z(kL) \otimes \mathcal{J}_k).$$

Since $kL - K_Z - kL = -K_Z$ is ample, Lemma 2.17(2) with $L' = kL$ implies

$$H^1(Z, \mathcal{O}_Z(kL) \otimes \mathcal{J}_k) = 0.$$

By the exact sequence

$$0 \to \mathcal{O}_Z(kL) \otimes \mathcal{J}_k \to \mathcal{O}_Z(kL) \to \mathcal{O}_Z(kL) \otimes (\mathcal{O}_Z/\mathcal{J}_k) \to 0,$$

we have

$$h^0(Z, \mathcal{O}_Z(kL) \otimes (\mathcal{O}_Z/\mathcal{J}_k)) = h^0(Z, \mathcal{O}_Z(kL)) - h^0(Z, \mathcal{O}_Z(kL) \otimes \mathcal{J}_k) \le h^0(Z, \mathcal{O}_Z(kL)) - \dim V_k.$$

Since q is an isolated point of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)$, the sheaf $\mathcal{O}_Z(kL) \otimes (\mathcal{O}_Z/\mathcal{J}_k)$ is the direct sum of a sheaf $\mathcal{F}_q$ supported at q with $h^0(Z, \mathcal{F}_q) = \ell(\mathcal{O}_{Z,q}/(\mathcal{J}_k)_q)$ and a sheaf supported on the closure of $\operatorname{Supp}(\mathcal{O}_Z/\mathcal{J}_k)\backslash\{q\}$. Hence

$$\ell(\mathcal{O}_{Z,q}/\mathfrak{a}_{t_k}(v)) \le \ell(\mathcal{O}_{Z,q}/(\mathcal{J}_k)_q) \le h^0(Z, \mathcal{O}_Z(kL)) - \dim V_k.$$

By asymptotic Riemann–Roch [Laz04a, Corollary 1.4.41],

$$\lim_{k\to+\infty} \frac{d! h^0(Z, \mathcal{O}_Z(kL))}{k^d} = (L^d) = \frac{I^d d^d (r+1)}{r}.$$

By Lemma 5.4,

$$\lim_{k\to+\infty} \frac{d! \dim V_k}{k^d} = I^d d^d.$$

Since $\lim_{k\to+\infty} t_k/k = I\sigma(v) > 0$, we have $t_k > 0$ for any sufficiently large k, and Definition 2.13 implies

$$I^d \sigma(v)^d \operatorname{vol}(v) = \lim_{k\to+\infty} \frac{d!\ell(\mathcal{O}_{Z,q}/\mathfrak{a}_{t_k}(v))}{t_k^d} \cdot \left(\frac{t_k}{k}\right)^d \le \frac{I^d d^d (r+1)}{r} - I^d d^d = \frac{I^d d^d}{r}.$$

Dividing by I^d, we obtain $\sigma(v)^d \operatorname{vol}(v) \le d^d/r$. □

6. Fano type over weighted hypersurfaces

In this section, we prove the following theorem.

Theorem 6.1. *Notation and conditions as in Set-up 4.1. Assume that Z is klt. Let Y be a $\mathbb{Q}$-factorial terminal projective variety of dimension d such that $\epsilon_{\mathrm{mov}}(-K_Y) = d$ and $\operatorname{vol}(-K_Y) = d^d$, and let $g : Y \dashrightarrow Z$ be a birational map such that g^{-1} does not contract any divisor. Then Y is of Fano type.*

We prove Theorem 6.1 at the end of this section, and the proof is divided into two cases according to the threshold τ of Construction 5.2. If τ is larger than one, then a general member of the anti-canonical sections of Y in a fixed degree, divided by the degree, is a klt boundary on Y (Lemma 6.2). If τ is at most one, then the bound of Proposition 5.3 singles out the valuation $\operatorname{ord}_\Gamma$, after a change of coordinates by Lemma 4.5 if r equals one (Proposition 6.3 and Lemma 4.3). We then extract Γ as a divisor on a variety over Y and reduce to Theorem 2.11 (Lemma 6.6).

Lemma 6.2. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Assume that $\tau > 1$. Then Y is of Fano type.*

Proof. By Lemma 5.4, $\mathfrak{b}_m \neq 0$ for any $m \geq 1$. Since $\tau > 1$, which includes the case $\tau = +\infty$, the definition of $\operatorname{lct}(Z; \mathfrak{b}_\bullet)$ in Definition 2.14 implies that there exists a positive integer m such that $mI \operatorname{lct}(Z; \mathfrak{b}_m) > 1$. Since $\mathfrak{b}_m^2 \subset \mathfrak{b}_{2m}$ and $\operatorname{mult}_P(\mathfrak{b}_m^2) = 2 \operatorname{mult}_P \mathfrak{b}_m$ for any prime divisor P over Z, we have

$$2m \operatorname{lct}(Z; \mathfrak{b}_{2m}) \geq 2m \operatorname{lct}(Z; \mathfrak{b}_m^2) = m \operatorname{lct}(Z; \mathfrak{b}_m).$$

Possibly replacing m with $2m$, we may assume that $mI \geq 2$.

Since IK_Z is Cartier, V_m is a non-zero linear subspace of $H^0(Z, \mathcal{O}_Z(-mIK_Z))$ with base ideal $\mathfrak{b}_m$, and

$$0 < \frac{1}{mI} < 1, \quad \frac{1}{mI} < \operatorname{lct}(Z; \mathfrak{b}_m).$$

By Lemma 2.18, there exists a member G of $|V_m|$ such that $(Z, G/(mI))$ is klt. Write $G = \operatorname{div}_Z(u) - mIK_Z$ with $u \in V_m = H^0(Y, \mathcal{O}_Y(-mIK_Y))$, and let $G_Y := \operatorname{div}_Y(u) - mIK_Y$. Then $G_Y \geq 0$, $G_Y \sim -mIK_Y$, and

$$K_Y + \frac{1}{mI} G_Y = \frac{1}{mI} \operatorname{div}_Y(u), \quad K_Z + \frac{1}{mI} G = \frac{1}{mI} \operatorname{div}_Z(u).$$

Let W be a smooth projective variety and $h : W \to Y$ a projective birational morphism such that $g \circ h : W \to Z$ is a morphism, which exists by resolving the closure of the graph of g [KM98, Theorem 0.2], and let Δ_W be the $\mathbb{Q}$-divisor on W such that

$$K_W + \Delta_W = \frac{1}{mI} \operatorname{div}_W(u) = h^* \Big(K_Y + \frac{1}{mI} G_Y \Big) = (g \circ h)^* \Big(K_Z + \frac{1}{mI} G \Big).$$

Then $h_* \Delta_W = G_Y/(mI)$ and $(g \circ h)_* \Delta_W = G/(mI)$. By [KM98, Lemma 2.30] applied to h and to $g \circ h$,

$$a\Big(P, Y, \frac{1}{mI} G_Y\Big) = a(P, W, \Delta_W) = a\Big(P, Z, \frac{1}{mI} G\Big) > 0$$

for any prime divisor P over Y. Thus $(Y, G_Y/(mI))$ is klt.

Since $\operatorname{vol}(-K_Y) = d^d > 0$, the Cartier divisor $-IK_Y$ is big. By Kodaira's lemma [Laz04a, Corollary 2.2.7] applied to $-IK_Y$, there exist an ample $\mathbb{Q}$-divisor A and a $\mathbb{Q}$-divisor $E \geq 0$ on Y such that $-K_Y \sim_{\mathbb{Q}} A + E$. Since Y is $\mathbb{Q}$-factorial, E is $\mathbb{Q}$-Cartier. Let η be a rational number such that $0 < \eta \ll 1$, and let

$$\Delta := (1 - \eta) \frac{1}{mI} G_Y + \eta E \geq 0.$$

Since $K_Y + G_Y/(mI) \sim_{\mathbb{Q}} 0$, we have

$$-(K_Y + \Delta) = -(1 - \eta) \Big(K_Y + \frac{1}{mI} G_Y \Big) - \eta (K_Y + E) \sim_{\mathbb{Q}} \eta A,$$

which is ample. By [KM98, Corollary 2.35(2)], $(Y, G_Y/(mI) + \eta E)$ is klt. By [KM98, Corollary 2.35(1)] applied to the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\eta G_Y/(mI) \geq 0$, (Y, Δ) is klt. Thus Y is of Fano type (Definition 2.2). □

The next proposition combines the upper bound in Proposition 5.3 with the lower bounds for normalized volumes in Lemma 4.3. When τ is at most one, Proposition 6.3 shows the following. If r is at least two, then the center of a valuation computing $\operatorname{lct}(Z; \mathfrak{b}_\bullet)$ is o, and both bounds are equalities. If r equals one, then the center is a smooth point, and the lower bound at the center is an equality.

Proposition 6.3. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Assume that $\tau \leq 1$. Then $\tau = 1$, and there exists $v \in \operatorname{Val}_Z$ computing $\operatorname{lct}(Z; \mathfrak{b}_\bullet)$. Moreover, for any $v \in \operatorname{Val}_Z$ computing $\operatorname{lct}(Z; \mathfrak{b}_\bullet)$, we have $\sigma(v) = a(v, Z)$, and the following hold.*

(1) If $r \geq 2$, then $v = c \operatorname{ord}_\Gamma$ for some real number $c > 0$.
(2) If $r = 1$, then $v = c \operatorname{ord}_q$ for some real number $c > 0$ and some closed point $q \in Z_{\mathrm{reg}}$.

Proof. Since $\tau \leq 1$, $\operatorname{lct}(Z;\mathfrak{b}_\bullet) < +\infty$. By Theorem 2.15(2), there exists $v \in \operatorname{Val}_Z$ computing $\operatorname{lct}(Z;\mathfrak{b}_\bullet)$.

Let $v \in \operatorname{Val}_Z$ be any valuation computing $\operatorname{lct}(Z;\mathfrak{b}_\bullet)$. By Definition 2.14, $a(v,Z) < +\infty$ and $\operatorname{lct}(Z;\mathfrak{b}_\bullet) = a(v,Z)/v(\mathfrak{b}_\bullet)$. Since $\operatorname{lct}(Z;\mathfrak{b}_\bullet) < +\infty$ and $a(v,Z)/0 = +\infty$, we have $v(\mathfrak{b}_\bullet) > 0$. Thus $\sigma(v) > 0$ and

$$a(v,Z) = I \operatorname{lct}(Z;\mathfrak{b}_\bullet) \cdot \frac{v(\mathfrak{b}_\bullet)}{I} = \tau\sigma(v).$$

By Lemma 5.5, $q := \operatorname{center}_Z v$ is a closed point of $Z_{\mathrm{reg}} \cup \{o\}$. Since $a(v,Z) < +\infty$ and $\sigma(v) > 0$, Proposition 5.3 gives $\sigma(v)^d \operatorname{vol}(v) \leq d^d/r$. Since $0 \leq \tau \leq 1$, we obtain

$$\widehat{\operatorname{vol}}(v) = a(v,Z)^d \operatorname{vol}(v) = \tau^d \sigma(v)^d \operatorname{vol}(v) \leq \tau^d \frac{d^d}{r} \leq \frac{d^d}{r}. \tag{6.1}$$

Suppose that $r \geq 2$. If $q \in Z_{\mathrm{reg}}$, then Lemma 4.3(1) implies that $\widehat{\operatorname{vol}}(v) \geq d^d > d^d/r$, which contradicts (6.1). Thus $q = o$. By Lemma 4.3(2), $\widehat{\operatorname{vol}}(v) \geq d^d/r$. By (6.1), $\tau^d \geq 1$ and $\widehat{\operatorname{vol}}(v) = d^d/r$. Thus $\tau = 1$, and the equality case of Lemma 4.3(2) implies that $v = c\operatorname{ord}_\Gamma$ for some real number $c > 0$.

Suppose that $r = 1$. By Lemma 4.2(3), $o \in Z_{\mathrm{reg}}$, so $q \in Z_{\mathrm{reg}}$. By Lemma 4.3(1), $\widehat{\operatorname{vol}}(v) \geq d^d$. By (6.1) with $r = 1$, $\tau^d \geq 1$ and $\widehat{\operatorname{vol}}(v) = d^d$. Thus $\tau = 1$, and the equality case of Lemma 4.3(1) implies that $v = c\operatorname{ord}_q$ for some real number $c > 0$.

In both cases $\tau = 1$, and hence $\sigma(v) = a(v,Z)$. □

The next lemma compares log discrepancies and asymptotic orders on Y and on Z.

Lemma 6.4. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Let W be a smooth projective variety and $h : W \to Y$ a projective birational morphism such that $g \circ h : W \to Z$ is a morphism, and let $F := h^*K_Y - (g\circ h)^*K_Z$. Then for any $v \in \operatorname{Val}_Z$, we have $a(v,Y) < +\infty$ if and only if $a(v,Z) < +\infty$, and in this case*

$$a(v,Y) = a(v,Z) - v(F), \quad \sigma_Y(v) = \sigma(v) - v(F).$$

Proof. Let $v \in \operatorname{Val}_Z$. Since W and Y are projective, v has a center on W and on Y. Since Y is terminal and Z is klt, $a(v,Y)$ and $a(v,Z)$ are defined (Definition 2.12(3)). By [BJ20, Section 1.5] applied to h and to $g \circ h$,

$$a(v,Y) = a(v,W) + v(K_W - h^*K_Y), \quad a(v,Z) = a(v,W) + v(K_W - (g\circ h)^*K_Z).$$

The second terms on the right-hand sides are real numbers. Thus $a(v,Y) < +\infty$ if and only if $a(v,W) < +\infty$, if and only if $a(v,Z) < +\infty$. In this case, subtracting the second equality from the first, we obtain

$$a(v,Y) - a(v,Z) = v\big((g\circ h)^*K_Z - h^*K_Y\big) = -v(F).$$

Let m be a positive integer. By Lemma 5.4, $V_m \neq 0$. For any non-zero $u \in V_m$, where $V_m = H^0(Y, \mathcal{O}_Y(-mIK_Y))$ (Construction 5.2), the divisors $\operatorname{div}_Y(u) - mIK_Y \geq 0$ and $\operatorname{div}_Z(u) - mIK_Z \geq 0$ are Cartier, and

$$h^*(\operatorname{div}_Y(u) - mIK_Y) = \operatorname{div}_W(u) - mIh^*K_Y = (g\circ h)^*(\operatorname{div}_Z(u) - mIK_Z) - mIF.$$

Since the ideals $\mathfrak{c}_m$ and $\mathfrak{b}_m$ are generated near the centers of v by local equations of the divisors $\operatorname{div}_Y(u) - mIK_Y$ and $\operatorname{div}_Z(u) - mIK_Z$ respectively, where u runs through the non-zero elements of V_m, we have

$$v(\mathfrak{c}_m) = \min_u v(\operatorname{div}_Y(u) - mIK_Y) = \min_u v(\operatorname{div}_Z(u) - mIK_Z) - mIv(F) = v(\mathfrak{b}_m) - mIv(F).$$

Dividing by mI and taking the infimum over m (Definition 2.14), we obtain

$$\sigma_Y(v) = \sigma(v) - v(F).$$

□

Remark 6.5. The birational morphism to X given by [BCHM10, Corollary 1.4.3] for a log canonical pair (X, Δ) is projective. The statement of [BCHM10, Corollary 1.4.3] does not include the projectivity, but the proof of [BCHM10, Corollary 1.4.3] constructs the domain of the morphism as a log terminal model over X, and a log terminal model over X is projective over X by [BCHM10, Definition 3.6.7].

If the center of ord_Γ on Y is not a divisor, then the next lemma extracts Γ over Y. An arbitrary extraction of Γ may lose the anti-canonical sections that do not vanish enough along Γ. Under the assumptions of the next lemma, the asymptotic order of $\mathfrak{c}_\bullet$ along Γ, divided by I, equals the log discrepancy of Γ over Y (Lemma 6.4). As a consequence, no anti-canonical section of Y is lost, and there is a klt pair on Y as needed in [BCHM10, Corollary 1.4.3].

Lemma 6.6. *Notation and conditions as in Set-up 5.1 and Construction 5.2. Assume that $\tau = 1$ and $\sigma(\mathrm{ord}_\Gamma) = a(\Gamma, Z)$. Then Y is of Fano type.*

Proof. **Step 1.** In this step, we show that $a(v, Y) \geq \sigma_Y(v)$ for any $v \in \mathrm{Val}_Z$ such that $a(v, Y) < +\infty$, that $\sigma_Y(\mathrm{ord}_\Gamma) = a(\Gamma, Y)$, and that $I \,\mathrm{lct}(Y; \mathfrak{c}_\bullet) \geq 1$.

Let $v \in \mathrm{Val}_Z$ be non-trivial. Since $\tau = 1$, $\mathrm{lct}(Z; \mathfrak{b}_\bullet) = 1/I$. By Theorem 2.15(1), $a(v, Z)/v(\mathfrak{b}_\bullet) \geq 1/I$. If $v(\mathfrak{b}_\bullet) > 0$, then $a(v, Z) \geq v(\mathfrak{b}_\bullet)/I = \sigma(v)$. If $v(\mathfrak{b}_\bullet) = 0$, then $a(v, Z) \geq 0 = \sigma(v)$. Thus

$$a(v, Z) \geq \sigma(v) \quad \text{for any nontrivial } v \in \mathrm{Val}_Z\,. \tag{6.2}$$

Let W be a smooth projective variety and $h : W \to Y$ a projective birational morphism such that $g \circ h$ is a morphism, which exists by resolving the closure of the graph of g [KM98, Theorem 0.2], and let $F := h^*K_Y - (g \circ h)^*K_Z$. Let $v \in \mathrm{Val}_Z$ be non-trivial such that $a(v, Y) < +\infty$. By Lemma 6.4, $a(v, Z) < +\infty$, and by (6.2),

$$a(v, Y) = a(v, Z) - v(F) \geq \sigma(v) - v(F) = \sigma_Y(v).$$

By Lemma 4.3, $a(\Gamma, Z) = d/r < +\infty$. By Lemma 6.4 and the assumption $\sigma(\mathrm{ord}_\Gamma) = a(\Gamma, Z)$,

$$a(\Gamma, Y) = a(\Gamma, Z) - \mathrm{mult}_\Gamma F = \sigma(\mathrm{ord}_\Gamma) - \mathrm{mult}_\Gamma F = \sigma_Y(\mathrm{ord}_\Gamma). \tag{6.3}$$

Let $m \geq 1$ and let $u \in H^0(Y, \mathcal{O}_Y(-mIK_Y))$ be non-zero. Since a local equation of $\mathrm{div}_Y(u) - mIK_Y$ at the generic point of $\mathrm{center}_Y(\mathrm{ord}_\Gamma)$ is contained in $\mathfrak{c}_m$, Definition 2.14 and (6.3) imply that

$$\mathrm{mult}_\Gamma(\mathrm{div}_Y(u) - mIK_Y) \geq \mathrm{mult}_\Gamma \mathfrak{c}_m \geq m \,\mathrm{ord}_\Gamma(\mathfrak{c}_\bullet) = mI\sigma_Y(\mathrm{ord}_\Gamma) = mIa(\Gamma, Y). \tag{6.4}$$

By Construction 5.2, $\mathfrak{c}_\bullet$ is a graded sequence of ideals on Y. Since Y is $\mathbb{Q}$-factorial and terminal, Y is a normal projective klt variety. For any non-trivial $v \in \mathrm{Val}_Y = \mathrm{Val}_Z$ such that $a(v, Y) < +\infty$ and $v(\mathfrak{c}_\bullet) > 0$, we have

$$\frac{a(v, Y)}{v(\mathfrak{c}_\bullet)} = \frac{a(v, Y)}{I\sigma_Y(v)} \geq \frac{1}{I}.$$

If $a(v, Y) = +\infty$ or $v(\mathfrak{c}_\bullet) = 0$, then $a(v, Y)/v(\mathfrak{c}_\bullet) = +\infty$. By Theorem 2.15(1), $I \,\mathrm{lct}(Y; \mathfrak{c}_\bullet) \geq 1$.

Step 2. In this step, we treat the case when $\mathrm{center}_Y(\mathrm{ord}_\Gamma)$ is a prime divisor on Y.

Suppose that $\mathrm{center}_Y(\mathrm{ord}_\Gamma)$ is a prime divisor on Y. By Lemma 4.4 applied to $g : Y \dashrightarrow Z$, $(\pi \circ g)^{-1}$ does not contract any divisor. Since $d \geq 3$ and $\epsilon_{\mathrm{mov}}(-K_Y) = d$, Theorem 2.11 applied to $\pi \circ g : Y \dashrightarrow \mathbb{P}^d$ implies that Y is of Fano type.

In the rest of the proof, we suppose that $\mathrm{center}_Y(\mathrm{ord}_\Gamma)$ is not a divisor on Y.

Step 3. In this step, we construct a projective birational morphism $\rho : Y' \to Y$ from a $\mathbb{Q}$-factorial normal projective variety Y' whose only exceptional prime divisor is Γ.

Since $\mathrm{center}_Y(\mathrm{ord}_\Gamma)$ is not a divisor and Y is terminal, $a(\Gamma, Y) > 1$. Let c be a rational number such that

$$1 - \frac{1}{a(\Gamma, Y)} < c < 1.$$

By **Step 1**, $I\operatorname{lct}(Y;\mathfrak{c}_\bullet) \geq 1 > c$. Since $\operatorname{lct}(Y;\mathfrak{c}_\bullet) = \sup_{m\geq 1} m\operatorname{lct}(Y;\mathfrak{c}_m)$ (Definition 2.14), there exists a positive integer m such that $c/(mI) < \operatorname{lct}(Y;\mathfrak{c}_m)$. Since $H^0(Y,\mathcal{O}_Y(-IK_Y)) \neq 0$, $H^0(Y,\mathcal{O}_Y(-mIK_Y)) \neq 0$. Since $0 < c/(mI) < 1$, Lemma 2.18 applied to the Cartier divisor $-mIK_Y$ and $V = H^0(Y,\mathcal{O}_Y(-mIK_Y))$ implies that there exists a member G of $|-mIK_Y|$ such that $(Y, cG/(mI))$ is klt. Let $D := G/(mI)$. Then $D \sim_{\mathbb{Q}} -K_Y$ and (Y,cD) is klt. Since $G = \operatorname{div}_Y(u) - mIK_Y$ for some non-zero $u \in H^0(Y,\mathcal{O}_Y(-mIK_Y))$, (6.4) implies $\operatorname{mult}_\Gamma G \geq mIa(\Gamma,Y)$. Thus $\operatorname{mult}_\Gamma D \geq a(\Gamma,Y)$, and

$$a(\Gamma,Y,cD) = a(\Gamma,Y) - c\operatorname{mult}_\Gamma D \leq (1-c)a(\Gamma,Y) < 1.$$

Let $h' : W' \to Y$ be a log resolution of (Y,cD) such that Γ is a divisor on W', which exists by [KM98, Lemma 2.45 and Theorem 0.2]. Since $\operatorname{center}_Y(\operatorname{ord}_\Gamma)$ is not a divisor, Γ is h'-exceptional. We check the conditions of [BCHM10, Corollary 1.4.3] for the klt pair (Y,cD), the log resolution h', and the set $\{\operatorname{ord}_\Gamma\}$. Condition (1) of [BCHM10, Corollary 1.4.3], that every element of $\{\operatorname{ord}_\Gamma\}$ has log discrepancy at most one with respect to (Y,cD), holds since $a(\Gamma,Y,cD) < 1$. Condition (2) there, that the center of every element of log discrepancy one does not contain any non-klt center, is vacuous since $a(\Gamma,Y,cD) \neq 1$. Thus, by [BCHM10, Corollary 1.4.3], where the hypothesis on Δ_0 there holds with $\Delta_0 = cD$, there exists a birational morphism $\rho : Y' \to Y$ such that Y' is $\mathbb{Q}$-factorial and the only ρ-exceptional prime divisor $\Gamma_{Y'}$ satisfies $\operatorname{ord}_{\Gamma_{Y'}} = \operatorname{ord}_\Gamma$. Moreover, ρ is projective by Remark 6.5. Thus Y' is a normal projective variety. Since $\Gamma_{Y'}$ is the only ρ-exceptional prime divisor,

$$K_{Y'} = \rho^* K_Y + (a(\Gamma,Y)-1)\Gamma_{Y'}. \tag{6.5}$$

Step 4. In this step, we show that $H^0(Y',\mathcal{O}_{Y'}(-kK_{Y'})) = H^0(Y,\mathcal{O}_Y(-kK_Y))$ as subspaces of $\mathbb{C}(Y)$ for any positive integer k divisible by I.

Since ρ is a morphism, ρ^{-1} does not contract any divisor. By Lemma 2.10(1) applied to $\rho : Y' \to Y$, $H^0(Y',\mathcal{O}_{Y'}(-kK_{Y'})) \subset H^0(Y,\mathcal{O}_Y(-kK_Y))$.

Conversely, write $k = mI$ and let $u \in H^0(Y,\mathcal{O}_Y(-kK_Y))$ be non-zero. Then $\operatorname{div}_Y(u) - kK_Y \geq 0$ is a Cartier divisor, and by (6.5),

$$\operatorname{div}_{Y'}(u) - kK_{Y'} = \rho^*(\operatorname{div}_Y(u) - kK_Y) - k(a(\Gamma,Y)-1)\Gamma_{Y'}.$$

For any prime divisor P on Y' other than $\Gamma_{Y'}$, the coefficient of $\operatorname{div}_{Y'}(u) - kK_{Y'}$ along P equals the coefficient of $\rho^*(\operatorname{div}_Y(u) - kK_Y)$ along P, which is non-negative. By (6.4), $\operatorname{mult}_\Gamma(\operatorname{div}_Y(u) - kK_Y) \geq ka(\Gamma,Y)$. Thus the coefficient of $\operatorname{div}_{Y'}(u) - kK_{Y'}$ along $\Gamma_{Y'}$ is

$$\operatorname{mult}_\Gamma(\operatorname{div}_Y(u) - kK_Y) - k(a(\Gamma,Y)-1) \geq k > 0.$$

Hence $\operatorname{div}_{Y'}(u) - kK_{Y'} \geq 0$, that is, $u \in H^0(Y',\mathcal{O}_{Y'}(-kK_{Y'}))$.

Step 5. In this step, we show that $\epsilon_{\mathrm{mov}}(-K_{Y'}) = d$.

Let $U_0 \subset Y'_{\mathrm{reg}}$ be a dense open subset such that $\rho|_{U_0}$ is an open immersion and $\rho(U_0) \subset Y_{\mathrm{reg}}$. Let $x \in U_0$ and let m be a positive integer. By **Step 4**,

$$H^0(Y',\mathcal{O}_{Y'}(-mIK_{Y'})) = H^0(Y,\mathcal{O}_Y(-mIK_Y)).$$

By Lemma 2.10(2) applied to ρ and U_0,

$$s(H^0(Y',\mathcal{O}_{Y'}(-mIK_{Y'})),x) = s(H^0(Y,\mathcal{O}_Y(-mIK_Y)),\rho(x)).$$

By Lemma 2.9 applied to the graded linear series

$$(H^0(Y',\mathcal{O}_{Y'}(-lK_{Y'})))_{l\geq 1} \quad \text{and} \quad (H^0(Y,\mathcal{O}_Y(-lK_Y)))_{l\geq 1},$$

we obtain

$$\begin{aligned} I\epsilon_{\mathrm{mov}}(-K_{Y'},x) &= \epsilon_{\mathrm{mov}}\big((H^0(Y',\mathcal{O}_{Y'}(-mIK_{Y'})))_{m\geq 1},x\big) \\ &= \epsilon_{\mathrm{mov}}\big((H^0(Y,\mathcal{O}_Y(-mIK_Y)))_{m\geq 1},\rho(x)\big) \\ &= I\epsilon_{\mathrm{mov}}(-K_Y,\rho(x)). \end{aligned}$$

Thus

$$\epsilon_{\mathrm{mov}}(-K_{Y'}, x) = \epsilon_{\mathrm{mov}}(-K_Y, \rho(x)) \quad \text{for any } x \in U_0. \tag{6.6}$$

Let $x \in U_0$ be a very general point of Y'. Since $\rho|_{U_0}$ is an open immersion, $\rho(x)$ is a very general point of Y. By Lemma 2.8 applied to $-K_{Y'}$ and to $-K_Y$ and by (6.6),

$$\epsilon_{\mathrm{mov}}(-K_{Y'}) = \epsilon_{\mathrm{mov}}(-K_{Y'}, x) = \epsilon_{\mathrm{mov}}(-K_Y, \rho(x)) = d.$$

Step 6. In this step, we conclude the proof.

Since $\mathrm{ord}_{\Gamma_{Y'}} = \mathrm{ord}_\Gamma$, $\mathrm{center}_{Y'}(\mathrm{ord}_\Gamma) = \Gamma_{Y'}$ is a prime divisor on Y'. The map $g \circ \rho : Y' \dashrightarrow Z$ is birational, and $(g \circ \rho)^{-1} = \rho^{-1} \circ g^{-1}$ does not contract any divisor since neither g^{-1} nor ρ^{-1} does. By Lemma 4.4 applied to $g \circ \rho : Y' \dashrightarrow Z$, $(\pi \circ g \circ \rho)^{-1}$ does not contract any divisor. Since $d \geq 3$ and $\epsilon_{\mathrm{mov}}(-K_{Y'}) = d$ (**Step 5**), Theorem 2.11 applied to $\pi \circ g \circ \rho : Y' \dashrightarrow \mathbb{P}^d$ implies that Y' is of Fano type. Since ρ^{-1} does not contract any divisor, Theorem 2.3 applied to $\rho : Y' \to Y$ implies that Y is of Fano type. □

Proof of Theorem 6.1. Since Y is $\mathbb{Q}$-factorial and Z is klt, K_Y and K_Z are $\mathbb{Q}$-Cartier. Since $\mathrm{vol}(-K_Y) = d^d > 0$, there exists a positive integer m_0 such that $H^0(Y, \mathcal{O}_Y(-m_0 K_Y)) \neq 0$. Let I_0 be a positive integer such that $I_0 K_Y$ and $I_0 K_Z$ are Cartier, and let $I := I_0 m_0$. Then IK_Y and IK_Z are Cartier, and $H^0(Y, \mathcal{O}_Y(-IK_Y)) \neq 0$ since it contains the I_0-th power of a non-zero element of $H^0(Y, \mathcal{O}_Y(-m_0 K_Y))$. Thus the conditions of Set-up 5.1 hold for Y, g, and I, and we use the notation of Construction 5.2.

If $\tau > 1$, then Y is of Fano type by Lemma 6.2. In the rest of the proof, we suppose that $\tau \leq 1$. By Proposition 6.3, $\tau = 1$, and there exists $v \in \mathrm{Val}_Z$ computing $\mathrm{lct}(Z; \mathfrak{b}_\bullet)$ such that $\sigma(v) = a(v, Z)$. By Lemma 6.6, it suffices to show that $\sigma(\mathrm{ord}_\Gamma) = a(\Gamma, Z)$. Note that $\sigma(cw) = c\sigma(w)$ for any $w \in \mathrm{Val}_Z$ and any real number $c > 0$, since $(cw)(\mathfrak{b}_m) = cw(\mathfrak{b}_m)$ for any $m \geq 1$.

Suppose that $r \geq 2$. By Proposition 6.3(1), $v = c\,\mathrm{ord}_\Gamma$ for some real number $c > 0$.

Suppose that $r = 1$. By Proposition 6.3(2), $v = c\,\mathrm{ord}_q$ for some real number $c > 0$ and some closed point $q \in Z_{\mathrm{reg}}$. By Lemma 4.5, there exist homogeneous coordinates $y_0, \ldots, y_{d+1}$ and a non-zero quadratic form $f' \in \mathbb{C}[y_1, \ldots, y_d]$ such that the conditions of Set-up 4.1 hold for $y_0, \ldots, y_{d+1}$ and f', and the point o defined in the coordinates $y_0, \ldots, y_{d+1}$ is q. Replacing $x_0, \ldots, x_{d+1}$ and f with $y_0, \ldots, y_{d+1}$ and f', we may assume that $q = o$. Then π and Γ denote the objects of Set-up 4.1 in the new coordinates, while the objects of Set-up 5.1 and Construction 5.2 do not depend on the coordinates.

By Lemma 4.2(3), $o \in Z_{\mathrm{reg}}$. By Lemma 4.3, Γ is the exceptional divisor of the blow-up of Z at o, so $\mathrm{ord}_\Gamma = \mathrm{ord}_o$ by Definition 2.12(4). Thus $v = c\,\mathrm{ord}_o = c\,\mathrm{ord}_\Gamma$.

In both cases, $v = c\,\mathrm{ord}_\Gamma$ for some real number $c > 0$. By Definition 2.12(3), $a(v, Z) = ca(\Gamma, Z)$. Since $c\sigma(\mathrm{ord}_\Gamma) = \sigma(v) = a(v, Z) = ca(\Gamma, Z)$, we have $\sigma(\mathrm{ord}_\Gamma) = a(\Gamma, Z)$. By Lemma 6.6, Y is of Fano type. □

7. Proof of the main theorem

In this section, we prove Theorem 7.1 and Theorem 1.2. We use [Zhu18, Proposition 1.9] to obtain a $\mathbb{Q}$-Fano variety Z, and we argue according to the classification of [LZ18, Theorem 3]. Only case (1) of Theorem 7.2 requires Theorem 6.1, and the other cases follow from results of [Zhu18].

Theorem 7.1. *Let Y be a $\mathbb{Q}$-factorial terminal projective variety of dimension $d \geq 3$ such that $\epsilon_{\mathrm{mov}}(-K_Y) = d$ and $\mathrm{vol}(-K_Y) = d^d$. Then Y is of Fano type.*

We use only one direction of [LZ18, Theorem 3], which we state in our notation.

Theorem 7.2 ([LZ18, Theorem 3])**.** *Let Z be a $\mathbb{Q}$-Fano variety of dimension d. Assume that there exists a smooth point $z \in Z$ such that $\epsilon(-K_Z, z) = d$. Then Z is isomorphic to one of the following varieties.*

(1) *A hypersurface* $(x_0x_{d+1} = f(x_1,\dots,x_d)) \subset \mathbb{P}(1^{d+1},r)$, *where* $r \geq 1$ *is an integer,* $x_0,\dots,x_d$ *are homogeneous coordinates of weight* 1, x_{d+1} *is a homogeneous coordinate of weight* r, *and* f *is a homogeneous polynomial of degree* $r+1$.
(2) *A hypersurface* $(x_{d+1}^2 + x_dh = f)$ *with* $h \neq 0$ *or a hypersurface* $(x_dx_{d+1} = f)$ *in* $\mathbb{P}(1^d,2,2)$, *where* $x_0,\dots,x_{d-1}$ *are homogeneous coordinates of weight* 1, x_d, x_{d+1} *are homogeneous coordinates of weight* 2, $h \in \mathbb{C}[x_0,\dots,x_{d-1}]$ *is homogeneous of degree* 2, *and* $f \in \mathbb{C}[x_0,\dots,x_{d-1}]$ *is homogeneous of degree* 4.
(3) *The blow-up of* $\mathbb{P}^d$ *along the complete intersection of a hyperplane and a hypersurface of degree at most* d.
(4) *The quotient of the quadric* $Q_k := (\sum_{i=0}^k x_i^2 = 0) \subset \mathbb{P}^{d+1}$ *by an involution* ι, *where* $2 \leq k \leq d+1$, $\iota^*x_i = \delta_i x_i$ *with* $\delta_i \in \{1,-1\}$ *for* $0 \leq i \leq d+1$, ι *is fixed point free in codimension one, and* $\delta_0,\dots,\delta_k$ *are not all equal.*
(5) *If* $d = 2$, *a Gorenstein log del Pezzo surface of degree at least* 4.

The Seshadri constant in [LZ18, Definition 1] is defined with ample in place of nef in Definition 2.7. For ample L, the value defined in [LZ18, Definition 1] equals $\epsilon(L,x)$, since the ample cone is the interior of the nef cone [Laz04a, Theorem 1.4.23(ii)].

Among the cases of Theorem 7.2, the next lemma singles out those in which the anti-canonical volume of Z equals the anti-canonical volume of Y in Theorem 7.1, so that Proposition 7.4 can be used. In case (1) the anti-canonical volume of Z is strictly larger (Lemma 4.2(4)), and this case is treated in Sections 4–6.

Lemma 7.3. *Let* Z *be a* $\mathbb{Q}$*-Fano variety of dimension* $d \geq 3$. *Assume that* Z *is isomorphic to a variety in Theorem 7.2(2) or Theorem 7.2(4). Then* $\mathrm{vol}(-K_Z) = d^d$.

Proof. Since $\mathrm{vol}(-K_Z)$ only depends on the isomorphism class of Z, replacing Z with a variety in Theorem 7.2(2) or Theorem 7.2(4), we may assume that Z is such a variety. Since Z is a $\mathbb{Q}$-Fano variety, Z is normal and K_Z is $\mathbb{Q}$-Cartier.

Case 1. Z is a hypersurface in Theorem 7.2(2). Let $\mathbb{P} := \mathbb{P}(1^d,2,2)$, let $S := \mathbb{C}[x_0,\dots,x_{d+1}]$ be the homogeneous coordinate ring of $\mathbb{P}$, with the weights in Theorem 7.2(2), let S_m be the degree m part of S, and let $F \in S_4$ be the defining polynomial of Z. Let $\ell := (x_0 = \cdots = x_{d-1} = 0) \subset \mathbb{P}$ and $\mathbb{P}^\circ := \mathbb{P}\backslash\ell$. For $0 \leq i \leq d-1$, the open subset $(x_i \neq 0)$ of $\mathbb{P}$ is isomorphic to $\mathbb{A}^{d+1}$ by [Dol82, 1.2.4], and $\mathcal{O}_{\mathbb{P}}(m)$ is invertible on $(x_i \neq 0)$ for any integer m: by [Dol82, Section 1.1], the sections of $\mathcal{O}_{\mathbb{P}}(m)$ over $(x_i \neq 0)$ form the $S_{(x_i)}$-module $S(m)_{(x_i)} = x_i^m S_{(x_i)}$, which is free with basis x_i^m. Thus $\mathbb{P}^\circ$ is smooth and $\mathcal{O}_{\mathbb{P}}(m)|_{\mathbb{P}^\circ}$ is invertible for any integer m. Since $\mathbb{P}$ is quasi-smooth, [Dol82, Theorem 3.3.4] implies that

$$\omega_{\mathbb{P}^\circ} \cong \mathcal{O}_{\mathbb{P}}(-d-4)|_{\mathbb{P}^\circ}.$$

Since Z is integral and $x_0 \notin FS$, the section x_0 of $\mathcal{O}_{\mathbb{P}}(1)$ does not vanish identically on $Z \cap \mathbb{P}^\circ$. Let H be the closure in Z of the divisor of zeros of $x_0|_{Z\cap\mathbb{P}^\circ}$. Then

$$\mathcal{O}_Z(mH)|_{Z\cap\mathbb{P}^\circ} \cong \mathcal{O}_{\mathbb{P}}(m)|_{Z\cap\mathbb{P}^\circ}$$

for any integer m. Let $V := Z_{\mathrm{reg}} \cap \mathbb{P}^\circ$. Since $\dim(Z \cap \ell) \leq 1 \leq d-2$ and Z is normal, we have $\mathrm{codim}_Z(Z\backslash V) \geq 2$. The variety V is a smooth prime divisor on the smooth variety $\mathbb{P}^\circ\backslash \operatorname{Sing} Z$ with ideal sheaf $F \cdot \mathcal{O}_{\mathbb{P}}(-4)|_{\mathbb{P}^\circ\backslash \operatorname{Sing} Z}$. By the adjunction formula [Har77, Proposition II.8.20],

$$\omega_V \cong \omega_{\mathbb{P}^\circ} \otimes \mathcal{O}_{\mathbb{P}}(4)|_V \cong \mathcal{O}_{\mathbb{P}}(-d)|_V \cong \mathcal{O}_Z(-dH)|_V.$$

Thus $K_Z|_V \sim -dH|_V$, and [Har77, Proposition II.6.5(b)] implies that $K_Z \sim -dH$.

For any integer j, the sheaf $\mathcal{O}_{\mathbb{P}}(2j)$ is invertible: by [Dol82, Section 1.1], its restriction to $(x_i \neq 0)$ is free with basis x_i^{2j} if $0 \leq i \leq d-1$, and with basis x_i^j if $i \in \{d, d+1\}$. For any integer j, the sheaves $\mathcal{O}_{\mathbb{P}}(2j)|_Z$ and $\mathcal{O}_Z(2jH)$ are isomorphic on $Z \cap \mathbb{P}^\circ$, so $\mathcal{O}_Z(2jH) \cong \mathcal{O}_{\mathbb{P}}(2j)|_Z$ by [Har77, Proposition II.6.5(b)]. In particular, $2H$ is a Cartier divisor on Z. Since $2dH \sim -2K_Z$ and $-K_Z$ is ample, $2H$ is ample. For any integer $j \geq 1$, there exists an exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}}(2j-4) \xrightarrow{\cdot F} \mathcal{O}_{\mathbb{P}}(2j) \to \mathcal{O}_{\mathbb{P}}(2j)|_Z \to 0.$$

By [Dol82, Theorem 1.4.1], $H^0(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m)) = S_m$ and $H^1(\mathbb{P}, \mathcal{O}_{\mathbb{P}}(m)) = 0$ for any integer m, as $\dim \mathbb{P} = d+1 \geq 4$. Thus

$$h^0(Z, \mathcal{O}_Z(2jH)) = \dim S_{2j} - \dim S_{2j-4},$$

which is the coefficient of t^{2j} in the power series

$$\frac{1-t^4}{(1-t)^d(1-t^2)^2} = \frac{1+t^2}{(1+t)(1-t)^{d+1}}.$$

The right-hand side has a pole of order $d+1$ at $t=1$ with

$$\lim_{t\to 1}(1-t)^{d+1} \cdot \frac{1+t^2}{(1+t)(1-t)^{d+1}} = 1,$$

and a pole of order 1 at $t=-1$. By partial fractions, the coefficient of t^m in the right-hand side is $m^d/d! + O(m^{d-1})$. Thus

$$h^0(Z, \mathcal{O}_Z(2jH)) = \frac{2^d}{d!} j^d + O(j^{d-1}).$$

On the other hand, [Laz04a, Corollary 1.4.41] implies that

$$h^0(Z, \mathcal{O}_Z(2jH)) = \frac{((2H)^d)}{d!} j^d + O(j^{d-1}).$$

Therefore, $((2H)^d) = 2^d$, that is, $(H^d) = 1$. Since $-2K_Z \sim 2dH$ is an ample Cartier divisor, [Laz04a, Corollary 1.4.41] implies that

$$\operatorname{vol}(-2K_Z) = ((2dH)^d) = (2d)^d (H^d) = (2d)^d,$$

and Theorem 2.6 implies that $\operatorname{vol}(-K_Z) = 2^{-d} \operatorname{vol}(-2K_Z) = d^d$.

Case 2. Z is the quotient of Q_k by ι in Theorem 7.2(4). Let $h : Q_k \to Z$ be the quotient morphism and H_Q a hyperplane section of $Q_k \subset \mathbb{P}^{d+1}$. Since $\delta_0, \dots, \delta_k$ are not all equal, there exist $0 \leq a, b \leq k$ with $\delta_a = 1$ and $\delta_b = -1$. Let $p \in \mathbb{P}^{d+1}$ be the point with $x_a = 1$, $x_b = \sqrt{-1}$, and $x_i = 0$ for $i \notin \{a, b\}$. Then $p \in Q_k$, and $\iota(p)$ has $x_a = 1$ and $x_b = -\sqrt{-1}$, so $\iota(p) \neq p$. Thus $\iota|_{Q_k} \neq \mathrm{id}_{Q_k}$, and h is finite of degree 2. The quadric Q_k has rank $k+1 \geq 3$, so Q_k is normal by [Har77, Exercise II.6.5(a)] applied to the affine cone over Q_k. By the adjunction formula [Har77, Proposition II.8.20] on $(Q_k)_{\mathrm{reg}}$ and [Har77, Proposition II.6.5(b)], we have $K_{Q_k} \sim -dH_Q$. Moreover, $(H_Q^d) = \deg Q_k$ by [Laz04a, Corollary 1.4.41], and $\deg Q_k = 2$ by [Har77, Proposition I.7.6(d)]. Since ι is fixed point free in codimension one and Z is normal, there exists an open subset $U \subset Z_{\mathrm{reg}}$ such that $\operatorname{codim}_Z(Z \backslash U) \geq 2$ and h is étale over U. Then $h^{-1}(U)$ is smooth and $K_{h^{-1}(U)} = (h|_{h^{-1}(U)})^* K_U$. Since h is finite, $\operatorname{codim}_{Q_k}(Q_k \backslash h^{-1}(U)) \geq 2$. Thus $K_{Q_k} = h^* K_Z$ by [Har77, Proposition II.6.5(b)].

Let N be a positive integer such that $-NK_Z$ is Cartier. By the projection formula [Ful98, Proposition 2.5(c)], applied d times, and since $h_*[Q_k] = 2[Z]$ by the definition of proper push-forward [Ful98, Section 1.4] as $\deg h = 2$,

$$2((-NK_Z)^d) = ((h^*(-NK_Z))^d) = ((-NK_{Q_k})^d) = N^d d^d (H_Q^d) = 2N^d d^d.$$

Thus $((-NK_Z)^d) = N^d d^d$. Since $-NK_Z$ is an ample Cartier divisor, [Laz04a, Corollary 1.4.41] implies that $\operatorname{vol}(-NK_Z) = ((-NK_Z)^d) = N^d d^d$, and Theorem 2.6 implies that $\operatorname{vol}(-K_Z) = N^{-d} \operatorname{vol}(-NK_Z) = d^d$. □

The proof of the next proposition follows [Zhu18, Remark 5.5] and [Zhu18, Lemma 2.9].

Proposition 7.4. *Let Y be a $\mathbb{Q}$-factorial terminal projective variety of dimension d such that $\operatorname{vol}(-K_Y) = d^d$, let Z be a $\mathbb{Q}$-Fano variety of dimension d such that $\operatorname{vol}(-K_Z) = d^d$, and let $g : Y \dashrightarrow Z$ be a birational map such that g^{-1} does not contract any divisor. Then $a(F, Z) \leq 1$ for any g-exceptional prime divisor F on Y, and Y is of Fano type.*

Proof. Suppose that $a(F,Z) > 1$ for some g-exceptional prime divisor F on Y. Let $t := a(F,Z) - 1 > 0$. Let W be a smooth projective variety and $h_Y : W \to Y$ a projective birational morphism such that $h := g \circ h_Y : W \to Z$ is a morphism. Such W and h_Y exist by resolving the closure of the graph of g [KM98, Theorem 0.2]. Let $F_W := (h_Y)_*^{-1}F$, and let N be a positive integer such that $-NK_Z$ is Cartier. By the definition of log discrepancies, t is the coefficient of F_W in $K_W - h^*K_Z$. Since $-NK_Z$ is Cartier, Nt is the coefficient of F_W in $NK_W - h^*(NK_Z)$, so $Nt \in \mathbb{Z}$, and NtF_W, which appears in (7.1), is a Weil divisor on W.

Let j be a positive integer and u a non-zero element of $H^0(Y, \mathcal{O}_Y(-NjK_Y))$. Then

$$D_Y := \operatorname{div}_Y(u) - NjK_Y \geq 0.$$

By Lemma 2.10(1), $u \in H^0(Z, \mathcal{O}_Z(-NjK_Z))$, that is,

$$D_Z := \operatorname{div}_Z(u) - NjK_Z \geq 0.$$

Since $-NK_Z$ is Cartier, D_Z is a Cartier divisor, and $h^*D_Z = \operatorname{div}_W(u) + jh^*(-NK_Z) \geq 0$. Since $\operatorname{mult}_{F_W} K_W = \operatorname{mult}_F K_Y$, we have $\operatorname{mult}_{F_W} h^*K_Z = \operatorname{mult}_F K_Y - t$. Thus

$$\operatorname{mult}_{F_W} h^*D_Z = \operatorname{ord}_F(u) - Nj(\operatorname{mult}_F K_Y - t) = \operatorname{mult}_F D_Y + Njt \geq Njt.$$

Since $\operatorname{mult}_E h^*D_Z \geq 0$ for any prime divisor $E \neq F_W$ on W, we have

$$\operatorname{div}_W(u) + j(h^*(-NK_Z) - NtF_W) = h^*D_Z - NjtF_W \geq 0.$$

Since j and u are arbitrary, we obtain an inclusion

$$H^0(Y, \mathcal{O}_Y(-NjK_Y)) \subset H^0(W, \mathcal{O}_W(j(h^*(-NK_Z) - NtF_W))) \tag{7.1}$$

of subspaces of $\mathbb{C}(Y) = \mathbb{C}(W)$ for any integer $j \geq 1$.

The morphism $h : W \to Z$ satisfies the condition in [Zhu18, Definition 2.7] (see the paragraph before Theorem 2.3), the line bundle $\mathcal{O}_Z(-NK_Z)$ is ample and hence big and nef, and $NtF_W \geq 0$ is a non-zero divisor on W. By [Zhu18, Lemma 2.15], with h, $\mathcal{O}_Z(-NK_Z)$, and NtF_W in the roles of the map, the line bundle, and the divisor, and since equality in [Zhu18, Lemma 2.15] holds only if the divisor is 0, we get $\operatorname{vol}(h^*(-NK_Z) - NtF_W) < \operatorname{vol}(-NK_Z)$. By Theorem 2.6 and (7.1),

$$N^d d^d = \operatorname{vol}(-NK_Y) \leq \operatorname{vol}(h^*(-NK_Z) - NtF_W) < \operatorname{vol}(-NK_Z) = N^d d^d,$$

a contradiction. Thus $a(F,Z) \leq 1$ for any g-exceptional prime divisor F on Y.

The birational map $g : Y \dashrightarrow Z$ satisfies the condition in [Zhu18, Definition 2.7]. In the notation of [Zhu18], the discrepancy of every g-exceptional prime divisor F on Y with respect to $(Z,0)$ is $a(F,Z) - 1 \leq 0$, and $(Z,0)$ is klt with $-K_Z$ ample since Z is a $\mathbb{Q}$-Fano variety. Thus [Zhu18, Lemma 2.9], with the map g, the pair $(Z,0)$, and the additional boundary 0, implies that Y is of Fano type. □

Proof of Theorem 7.1. Since Y is terminal, Y is klt ([KM98, Definition 2.34]). Since Y is a projective klt variety and $\epsilon_{\mathrm{mov}}(-K_Y) = d$, by [Zhu18, Proposition 1.9], there exist a birational map $g : Y \dashrightarrow Z$ to a $\mathbb{Q}$-Fano variety Z and a smooth point $z \in Z$ such that g satisfies the condition in [Zhu18, Definition 2.7], that is, g^{-1} does not contract any divisor, and $\epsilon(-K_Z, z) \geq d$. Here we use that, by [Zhu18, Definition 1.1], the Seshadri constant $\epsilon(-K_Z)$ of [Zhu18] is the maximum of $\epsilon(-K_Z, x)$ over smooth points $x \in Z$.

If $\epsilon(-K_Z, z) > d$, then $Z \cong \mathbb{P}^d$ by [LZ18, Theorem 2]. Let $\psi : Z \to \mathbb{P}^d$ be an isomorphism. Then $(\psi \circ g)^{-1} = g^{-1} \circ \psi^{-1}$ does not contract any divisor. Since $d \geq 3$ and $\epsilon_{\mathrm{mov}}(-K_Y) = d$, Y is of Fano type by Theorem 2.11. Thus we may assume that $\epsilon(-K_Z, z) = d$. By Theorem 7.2, there exists an isomorphism $\phi : Z \to Z_1$ onto a variety Z_1 in one of the cases (1)–(5) of Theorem 7.2. Then Z_1 is a $\mathbb{Q}$-Fano variety, and $(\phi \circ g)^{-1} = g^{-1} \circ \phi^{-1}$ does not contract any divisor. Replacing Z and g with Z_1 and $\phi \circ g$, we may assume that Z is a variety in one of the cases (1)–(5) of Theorem 7.2.

Case 1. Z is a hypersurface $(x_0x_{d+1} = f(x_1, \ldots, x_d)) \subset \mathbb{P}(1^{d+1}, r)$ in Theorem 7.2(1). If $f = 0$, then $Z = (x_0x_{d+1} = 0)$ is reducible, which is not possible as Z is a variety. Thus $f \neq 0$. Since

$d \geq 3$, $r \geq 1$, and Z is a $\mathbb{Q}$-Fano variety, Z is normal and klt, and the conditions of Set-up 4.1 hold (a Weil divisor H with $\mathcal{O}_Z(H) \cong \mathcal{O}_Z(1)$ exists, since $\mathcal{O}_Z(1)$ is a reflexive sheaf of rank one, as for $\mathcal{O}_Z(d\ell)$ in the proof of Lemma 5.5). By Theorem 6.1, Y is of Fano type.

Case 2. Z is a variety in Theorem 7.2(2). By Lemma 7.3, $\mathrm{vol}(-K_Z) = d^d$. By Proposition 7.4, Y is of Fano type.

Case 3. Z is a variety in Theorem 7.2(3). Then there exists a birational morphism $b : Z \to \mathbb{P}^d$. Since any prime divisor on $\mathbb{P}^d$ has a strict transform on Z, b^{-1} does not contract any divisor. Thus $(b \circ g)^{-1} = g^{-1} \circ b^{-1}$ does not contract any divisor. Since $d \geq 3$ and $\epsilon_{\mathrm{mov}}(-K_Y) = d$, Y is of Fano type by Theorem 2.11.

Case 4. Z is a variety in Theorem 7.2(4). By Lemma 7.3, $\mathrm{vol}(-K_Z) = d^d$. By Proposition 7.4, Y is of Fano type.

Case 5. Z is a variety in Theorem 7.2(5). Then $\dim Z = 2 < 3 \leq d = \dim Y$, which is not possible as g is birational. □

Proof of Theorem 1.2. Let X be a normal projective variety of dimension $d \geq 1$ such that $\epsilon_{\mathrm{mov}}(-K_X) \geq d$. If $\epsilon_{\mathrm{mov}}(-K_X) > d$, or $\epsilon_{\mathrm{mov}}(-K_X) = d$ and $\mathrm{vol}(-K_X) > d^d$, or $d \leq 2$, then X is of Fano type by Proposition 3.2. Thus we may assume that $d \geq 3$, $\epsilon_{\mathrm{mov}}(-K_X) = d$, and $\mathrm{vol}(-K_X) \leq d^d$. By Proposition 3.2,

$$\mathrm{vol}(-K_X) \geq \epsilon_{\mathrm{mov}}(-K_X)^d = d^d,$$

so $\mathrm{vol}(-K_X) = d^d$. Let $f : Y \to X$ be a projective birational morphism such that Y is $\mathbb{Q}$-factorial and terminal and K_Y is nef over X, which exists by Lemma 3.1. By Lemma 3.1(2), $\epsilon_{\mathrm{mov}}(-K_Y) = d$ and $\mathrm{vol}(-K_Y) = d^d$. By Theorem 7.1, Y is of Fano type. By Lemma 3.1(3), X is of Fano type. □

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